Scalable Reactive Atomistic Dynamics with GAIA

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Abstract

The groundbreaking advance in materials and chemical research has been driven by the development of atomistic simulations [1–6]. However, the broader applicability of the atomistic simulations remains restricted, as they inherently depend on energy models that are either inaccurate or computationally prohibitive. Machine learning interatomic potentials (MLIPs) have recently emerged as a promising class of energy models, but their deployment remains challenging due to the lack of systematic protocols for generating diverse training data. Here we automate the construction of training datasets to enable the development of general-purpose MLIPs, by introducing GAIA, an end-to-end framework to build a wide range of atomic arrangements. By employing systematic evaluation of metadynamics for effective structural exploration, GAIA overcomes the heuristic nature of conventional dataset generation. Using GAIA, we constructed Titan25, a benchmark-scale dataset, and trained MLIPs that closely match both static and dynamic density functional theory results. The models further reproduce experimental observations across reactive regimes, including detonation, coalescence, and catalytic activity. GAIA narrows the gap between experiment and simulation, and paves the way for the development of universal MLIPs that can reliably describe a wide spectrum of materials and chemical processes.

1 Intro

Atomistic simulations are advancing the frontier of materials and chemical sciences. This progress has been driven by simulation techniques such as molecular dynamics (MD) and Monte Carlo, which have been pivotal in elucidating phenomena that remain inaccessible to direct experimental observation [7–11]. However, the fidelity of these simulations depends on the underlying energy models, which involve inevitable trade-offs between predictive accuracy and computational efficiency. Classical force fields (FFs) are widely used energy models that approximate complex atomistic interactions with simple analytical functions, thereby enabling computational efficiency essential for large-scale simulations [12, 13]. Nevertheless, their reliance on heuristics and system-specific parameter fitting limits their general applicability, particularly for systems containing metallic elements [14]. This limitation is compounded as only a few FFs can describe bond formation and breaking, a capability essential for modeling complex chemical processes, thus further restricting their use to narrowly defined systems and applications [15–17]. Density functional theory (DFT) can address these challenges by providing an accurate description of interatomic interactions, but the computational cost increases significantly with system size, limiting its applicability to simplified systems rather than experimentally relevant ones.

Recently, machine learning interatomic potentials (MLIPs) have emerged as a promising approach for large-scale atomistic simulations [18]. They have attracted considerable attention due to the ability to achieve linear computational scaling, enabling simulations of large, experimentally relevant systems. By learning from high-quality ab initio reference data across a broad chemical space, MLIPs enable large-scale simulations at ab initio accuracy, thereby allowing simulation techniques to narrow the gap between computations and experiments [19, 20]. Conversely, in the absence of sufficient and diverse training data, the performance of MLIPs deteriorates, limiting their reliability. This limitation, together with the absence of a general-purpose MLIP applicable across diverse simulations with a single model, highlights the need for comprehensive protocols for generating a broad spectrum of representative atomic configurations.

In this paper, we introduce GAIA, a framework to generate datasets with an automated pipeline for machine learning inter-atomic potentials. GAIA is explicitly designed to overcome the challenges outlined above, namely, the lack of systematic strategies for sampling structural space and the difficulty of ensuring transferability in MLIPs, thereby enabling diverse simulations with a single MLIP model. To achieve this, GAIA employs an end-to-end process to generate a diverse ensemble of atomic configurations for MLIP training, starting from simple seed coordinate information.

Using the GAIA framework, we introduce Titan25, a representative dataset comprising 1.8 millions atomic configurations across 11 elements. Here, we show that an MLIP model trained on the Titan25 dataset, SNet-T25, achieves good agreement with DFT and experimental results, including metal–organic adsorption distances, coalescence of carbon nanotubes, detonation product distributions, and the water–gas shift reaction mechanism (Fig. 1a,b). We also demonstrate that the SNet-T25 provides more accurate and reliable energies, forces, and geometries than current state-of-the-art models on out-of-distribution datasets. This establishes GAIA as a general framework for developing transferable MLIPs across diverse chemical spaces.

2 Toward generalizable MLIPs

The generalization of MLIPs critically hinges on training data that encompass structural diversity [21]. In principle, a dataset that fully spans the chemical space would resolve this limitation, yet constructing such a dataset is infeasible, leaving generalization as a persistent challenge. Current state-of-the-art approaches remain constrained, as they either depend on incremental refinements of existing databases [22–26], limiting structural diversity, or incur substantial computational overhead beyond generation stages [27–29] (Supplementary Note 9.1). To address these limitations, we designed GAIA to generate datasets that broadly represent the accessible chemical space defined by user-chosen components, while minimizing redundancy and ensuring transferability across a wide range of chemical environments. This concept enables the generation of near- and non-equilibrium atomic structures across the targeted chemical space, with the dataset scaling systematically as the number of the chosen components increases. Titan25, a representative dataset generated by GAIA, comprises a moderate number of data points and elements (Fig. 1c) yet containing the well-balanced compositions without undue bias toward any single element (Fig. 1d). The per-atom energy distribution of Titan25 is shifted toward higher energy values compared to ANI-1xnr [29] and MPtrj [30], and the 95$^{\text{th}}$ percentile of atomic force norms is 7.1 eV/ Å, higher than in both datasets as well as in OMat24 [24] and OMol25 [25] (Fig. 1e and Supplementary Fig. 3d,e). These statistics indicate that Titan25 spans a broader range of energetic and force regimes, encompassing various near- and non-equilibrium structures. Such coverage provides informative training signals for MLIPs, particularly in reactive regimes, and positions Titan25 as a prototype dataset that can be further extended to broader chemical spaces.

3 GAIA framework

The GAIA framework aims to provide comprehensive training data that enable a single MLIP model to generalize across crystalline, amorphous, molecular, and interfacial systems. It consists of two main modules: the data-generator (DG), which produces a broad spectrum of atomic arrangements by introducing multiple physics-inspired structure builders, and the data-improver (DI), which augments them through coverage-enhancing sampling of structural space. The DG module generates DFT-labeled configurations from user-provided simple seed components—such as H_2O, CO_2, Pt(111)—using six builders: Checkerboard, Bulk, Slab, Adatom, Admol, and Nanoreactor$^+$ (Fig. 2a). The first five builders generate structures by varying the input seed structures according to predefined protocols. In contrast, Nanoreactor$^+$ facilitates the modeling of chemical transformation and produces previously inaccessible non-equilibrium configurations. Together, these builders generate periodic bulk, surface, mixed-composition, and reactive configurations, thereby expanding the diversity of the training data (Supplementary Fig. 1 and Methods 8.1). The clustered embedding observed in Fig. 2b highlights the complementary sampling patterns of different builders and shows that configurations generated by Nanoreactor$^+$ occupy regions of structural space distinct from those of the other builders. The DI module generates additional training data, targeting regions underrepresented in the DG stages or those with high model errors (Fig. 2c). It is a novel method that serves as a cost-efficient alternative to approaches primarily based on brute-force oversampling [31, 32] or those relying on the guidance of models trained earlier [27–29]. The identification of the improvable regions is performed by data categorization based on d_Λ, which quantifies diversity across atomic configurations (Method 8.3). Fig. 2d compares structures generated by the DG and DI modules in terms of total energy and d_Λ. The DI module extends sampling into regions scarcely populated by DG, as seen in the marginal distributions, thereby enhancing coverage of diverse configurations. Both DG and DI are fully automated and require no prior domain knowledge, providing a scalable alternative to conventional manual curation or other computationally costly approaches that have constrained the scope of MLIP datasets.

4 GAIA benchmark

To assess the ability of MLIP models to predict fundamental physical properties, we constructed GAIA-Bench, which benchmarks against DFT energies and forces across intermolecular interactions (mol2mol), bulk energy–volume relations (bulk), surface facet stability (slab), and molecule–surface adsorption energetics (mol2surf). In the development, the benchmark structures were deliberately selected to avoid overlap with the datasets under evaluation, allowing an unbiased assessment of generalization ability. For the force evaluation, perturbed configurations of the backbone structures were additionally considered at multiple levels (ϵ) to probe a wider range of force variations. Further details of GAIA-Bench can be found in Method 8.5. Using GAIA-Bench, we evaluated the performance of MLIPs trained on two datasets generated by GAIA, Titan25(G) (from DG only) and Titan25(G+I) (from both DG and DI), against two existing public datasets, ANI-1xnr and MPtrj. For a fair comparison, the same model architecture was used in training, with details provided in Method 8.4. Among the four MLIPs, one with Titan25(G+I) consistently yielded the lowest errors, with energy errors reduced by about a factor of two in average across the four tasks and force errors by about one-third compared with the public datasets, and with $\sim$20% lower values for the both metrics relative to Titan25(G) (Fig. 3a and Supplementary Fig. 6). In the forces evaluation, the area of errors on the public datasets increases drastically with respect to ϵ, while those on Titan25 maintain relatively stable (Fig. 3b). Moreover, Titan25(G+I) leads to the smallest errors also in both the magnitude and angular components of the force (Fig. 3c). These findings establish that training data from DG delivers consistently strong performance across the various configurations, which is further amplified through the integration with DI, paving the way toward broadly transferable MLIPs.

5 Reproducing prototypical surface reactions: Pt–CO_x–H_2O

Beyond the benchmark evaluation of static energy and force predictions of MLIPs, we further examine their ability to reproduce dynamical interfacial processes through MD simulations of CO and CO_2 adsorption on Pt(111) in aqueous solution [33–36].

In this evaluation, the SevenNet model [37] trained on the Titan25(G+I) dataset, termed SNet-T25, is compared with UMA [38], a massive model jointly trained on five large-scale MLIP datasets (OC20, ODAC, OMat, OMol, and OMC). The details of the simulation setups and the models are described in Methods 8.6. As shown in Fig. 3d, four models including SNet-T25, consistently reproduced the Pt–CO adsorption distance (∼1.4 Å), in close agreement with DFT. In contrast, UMA-OMol and -OMC, which were specialized for molecular data, failed to capture the stability of the Pt surface. For the CO_2 adsorption case, only SNet-T25 maintained the molecules close to the Pt surface (∼3.4 Å), whereas the UMA variants showed large fluctuations, indicative of repulsive interactions (Fig. 3e). We also performed DFT calculations on reduced CO_2–Pt systems, which provided the distance values consistent with SNet-T25. Overall, the results demonstrate that Titan25 leads to accurate representation of both the CO and CO_2 adsorption, outperforming the five-dataset collection. This highlights the ability of Titan25-trained MLIPs to reproduce realistic interfacial dynamics with near-ab initio fidelity, capturing the intricate behavior of molecules at solid-liquid boundaries under chemically complex environments, despite SNet-T25 being markedly smaller in model size and trained on considerably less data than UMA.

6 Validation across experimental phenomena

Beyond achieving DFT-level accuracy, the ultimate value of MLIPs lies in reproducing experimental observables. However, reproducing experimental phenomena for systems not represented in their training domain remains a challenge for MLIPs. In the following subsections we demonstrate that GAIA-based training enables MLIPs to generalize across diverse, experimentally relevant chemical processes with high reliability. We simulate detonation, coalescence, adsorption, and catalytic mechanisms, spanning from organic systems to metal-molecule interfacial processes using only a single Titan25-trained potential, SNet-T25. See Method 8.6 for detailed simulation conditions.

6.1 Detonation of high energetic molecules

Detonation is a shock-driven combustion process at extreme pressures, where accurate prediction of product distributions is essential for assessing energy release and validating models [39, 40]. We performed Hugoniostat simulations at 40 GPa for TNT, TATB, RDX, and nitromethane using SNet-T25, and compared the normalized product ratios with both the experimental data [41] (Fig. 4a) and the domain-specific MLIP, Gen3.9zbl [40]. As shown in Fig. 4b, SNet-T25 predicts the formation of H_2O, N_2, CO_2, and NH_3, with normalized ratios comparable to both the experiments and Gen3.9zbl for all the energetic molecules. Deviations were smallest for nitromethane, remaining below 4% across all products, and largest for TATB, reaching 18% for N_2 due to the overproduction. It also reproduced the experimental product distribution orderings across all systems, with the exception of TNT, for which the three major products were obtained in nearly identical ratios (within ∼2% deviation). Supplementary Fig. 9 shows that the model yields consistent ratios across multiple pressures, demonstrating that Titan25-trained MLIPs remain accurate and stable in high-pressure detonation regimes without additional correction terms, such as Ziegler–Biersack–Littmark (ZBL) potential [40, 42].

6.2 Coalescence of carbon nanotubes

The coalescence of carbon nanotubes (CNTs) provides a route to tune their diameter and morphology, but simulating this process has been difficult because DFT cannot reach the relevant scales and classical force fields struggle to capture bond rearrangements [43, 44]. Using Titan25, we simulated the coalescence of armchair CNTs with chiral indices (6,6), (9,9), (12,12), (18,18), and (24,24) (Fig. 4c-e). For the (6,6), (9,9), and (12,12) pairs, the resulting pores corresponded to their doubled tubes, (12,12), (18,18), and (24,24), in agreement with experimental observations. By contrast, (24,24) tubes collapsed into ribbon-like structures rather than forming (48,48) cylinders, also consistent with the experimental trend that large-diameter CNTs ($\approx 4.2 - 6.9$ nm) lose stability and flatten upon contact [45, 46]. These results shows that Titan25 reliably describes bond rearrangements in extended carbon systems, enabling predictive simulations of size- and morphology-dependent CNT behavior.

6.3 π-conjugated molecules on metal surfaces

The equilibrium adsorption distance of π-conjugated molecules on metals is a key descriptor of hybrid interface stability, yet its experimental determination remains restricted to highly ordered overlayers and requires complex analysis [47–54]. Fig. 5a,b compares benzene (BZ), diindenoperylene (DIP), and fullerene (C_{60}) on Ag(111) from the experiments [55], DFT [55], and SNet-T25. The SNet-T25 reproduces parallel adsorption geometries with average distance deviations of about 5% across metals, in close agreement with the experiments (Supplementary Fig. 11). For DIP, the deviation is slightly larger (8%), yet the model correctly captures the relative order of experimental adsorption distances Cu $<$ Ag $<$ Au. A more complex case is water encapsulated inside C_{60} on Ag(111) [56] (Fig. 5c), where SNet-T25 reproduces the experimental Ag–O distance (5.60 Å vs. 5.57 Å) during MD simulation as well.

Together, these results show that our model captures the delicate balance of dispersion, orbital hybridization, and intramolecular reorganization that governs π-conjugated molecules on metal surfaces, providing a physically grounded framework for interpreting and predicting interfacial stability.

6.4 Water-gas-shift reaction on Au(100)

The water–gas shift reaction ($CO + H_2O \rightarrow CO_2 + H_2$) is a key route to hydrogen production, yet its catalytic mechanism remains contested, as activity is highly complex and sensitive to subtle variations in reaction conditions [57, 58]. Using SNet-T25, we performed large-scale MD simulations of CO and H_2O on hydroxylated Au(100) (Fig. 5d). The trajectories capture the essential features of two pathways: in the carboxyl mechanism, CO adsorption is followed by formation of the OCOH* intermediate and subsequent CO_2 release by adjacent H_2O and OH; in the redox mechanism, H abstraction yields O*, which couples with CO* to form CO_2. These results demonstrate that Titan25 could be used not only to reproduce experimentally established reaction channels but also to resolve their dynamics under realistic catalytic conditions. This underscores its potential for predictive heterogeneous catalysis.

7 Conclusion

We demonstrated that GAIA systematically generates diverse and high-quality datasets, enabling MLIPs to capture not only static energies and forces but also dynamical processes across molecules, solids, interfaces, and chemical reactions. This is exemplified by Titan25, a benchmark-scale dataset comprising 1.8 million structures, which includes diverse chemical reactions and showcases the broad expressivity of GAIA-based datasets. A notable advantage is that GAIA eliminates the reliance on expert heuristics, which has traditionally been required to curate training datasets, thereby allowing automated dataset generation. This data-centric strategy also confers strong performance on out-of-distribution structures, as demonstrated by GAIA-Bench, where the models trained on GAIA-generated datasets retained accuracy even for perturbed and previously unseen configurations. The SNet-T25 model reproduced experimental observables ranging from detonation products to catalytic pathways, showing that the automated dataset generation can yield transferable MLIPs. These results suggest that MLIPs trained on GAIA data can evolve into predictive engines for exploring materials and chemical space under realistic conditions, including those that remain inaccessible to conventional simulations.

The configuration space of atomic systems is, in principle, infinite, raising the question of whether a truly single-model universal MLIP is achievable or remains an aspirational ideal. In practice, constructing such a model is hindered because there is no general way to identify which regions of chemical space are adequately sampled and which remain underrepresented, making the assembly of truly comprehensive datasets infeasible. Nevertheless, by systematically and automatically generating datasets initiated from given simple components and progressively exploring the surrounding regions of chemical space, GAIA provides a practical foundation for the development of universal MLIPs.

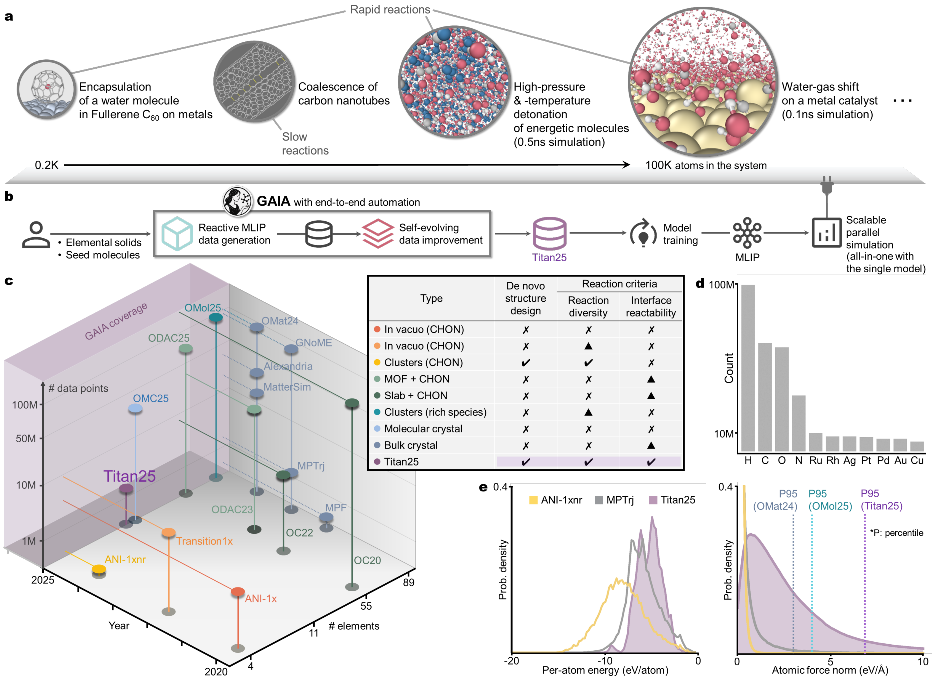

Fig. 1: **Large-scale atomistic simulations with the Titan25-trained MLIP.** **a**, MLIPs have traditionally lacked transferability across diverse reactions, making it difficult to apply broadly as a general model. Here, we show representative simulations spanning four scales, from left to right: molecular adsorption, nanoscale coalescence, non-equilibrium assembly, and catalytic reactions—demonstrating the broad applicability of a model trained on Titan25. **b**, Schematic overview of the GAIA framework for generating and applying MLIPs. From left to right, user-provided input structures are augmented by GAIA into diverse datasets (for example, Titan25) comprising a wide range of atomic arrangements. These datasets can then be trained with any third-party training framework to yield models, which are subsequently applied in the simulations. **c**, Comparison of dataset scales, with axes showing the released years, the number of included elements, and the number of snapshots. GAIA can generate datasets of arbitrary size and elemental diversity. As a representative case, Titan25 comprises 11 elements and 1.8 million snapshots. The rationale for the table is found in Supplementary Note 9.1. **d**, The number of atoms for each element contained in Titan25. **e**, Probability density distributions of per-atom energies (left panel) and norms of atomic forces (right panel). See Supplementary Fig. 3 for details.

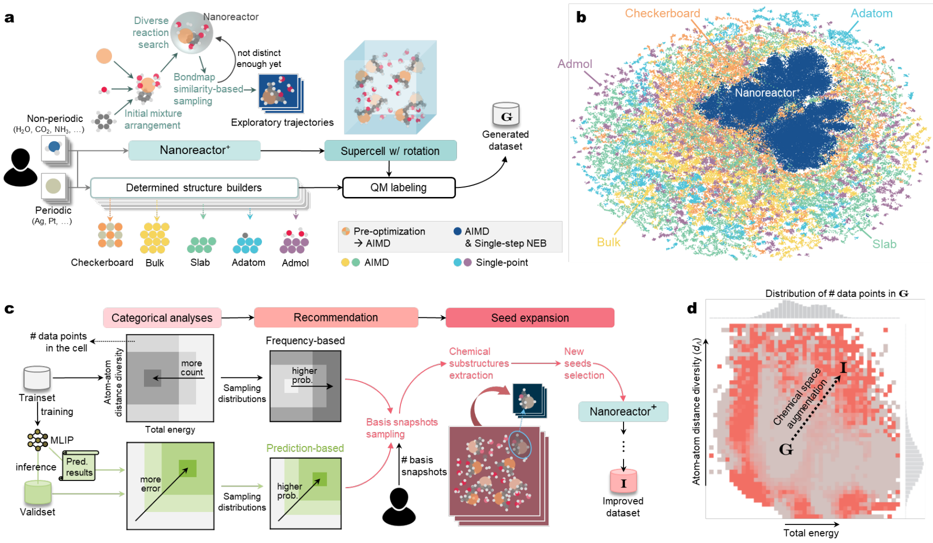

Fig. 2: **GAIA data-generator and -improver produce diverse datasets.** **a**, Illustration of DG module. Periodic and non-periodic inputs are expanded into diverse unlabeled configurations by builders such as Checkerboard, Bulk, Slab, Adatom, Admol, and Nanoreactor$^+$. The former builders generate structures by retaining key input features or combining structures, while Nanoreactor$^+$ explores reactive configurations by inducing bond formation and cleavage via enhanced dynamics. **b**, An embedding plot of feature vectors extracted from structures using an MLIP trained on the DG-generated dataset (dataset G). Each point corresponds to a generated structure, and the color denotes the builder from which the structure originated. **c**, Illustration of DI module. It extracts basis snapshots by performing categorical analyses, based on data point counts for the training set and error-based measures for the model predictions, respectively. New seed structures selected from these snapshots are fed into the Nanoreactor$^+$ builder, where they are expanded into diverse configurations to form dataset I. **d**, Comparison of structures generated by DG and DI using atom–atom distance diversity (d_Λ) and total energy metrics. Heatmap elements denote the relative dominance of each dataset by counts, with gray indicating dataset G and red indicating dataset I. The distribution of dataset G is further illustrated as a histogram on the side. Red regions correspond to dataset I filling sparse areas of dataset G, demonstrating its contribution of novel structures beyond DG.

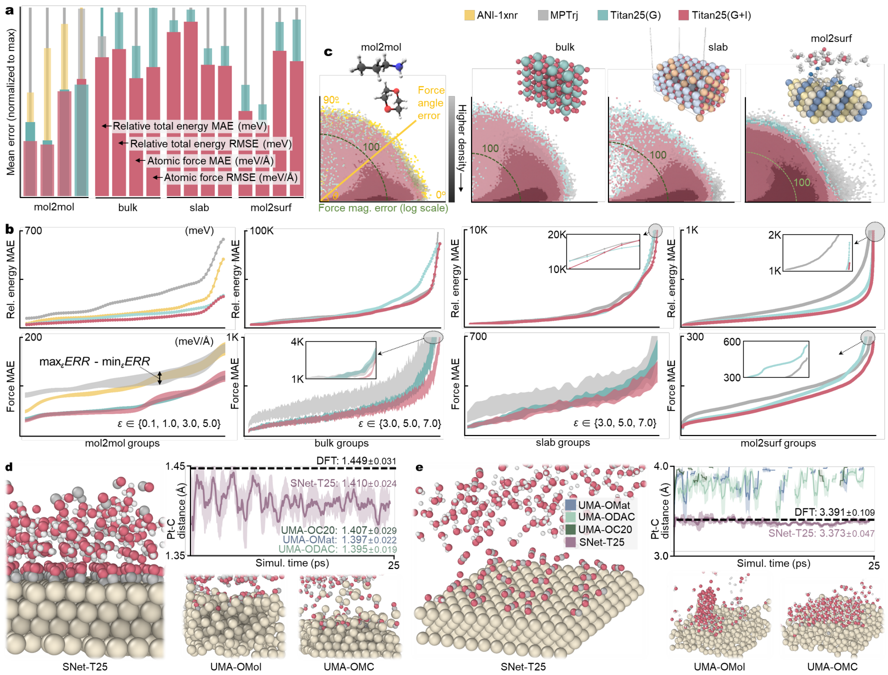

Fig. 3: **The Titan25-trained MLIP delivers reliable accuracy both in static and dynamic benchmarks. a**, Normalized mean error plot on GAIA-Bench for the static benchmark. The normalization ensures that all bars are scaled to a common maximum for relative comparison. ANI-1xnr is only plotted for mol2mol, as it does not include metallic elements. **b**, Mean absolute errors (MAEs) of relative energies (top) and forces (bottom) for the four GAIA-Bench tasks. Each plot is individually sorted in ascending order; the horizontal axis corresponds to each group having variant snapshots from the same reference structure. In the lower panel, shaded regions represent the range between the smallest and largest MAE values with respect to ϵ, structure perturbation strength at which the force norm matches the threshold. In the case of mol2surf, the perturbations were not introduced; instead, only the structures used for energetic calculations were considered. An inset provides an enlarged view of the plot near its end. Root-mean-square error (RMSE) results are provided in Supplementary Fig. 7. **c**, Angle and magnitude errors in atomic forces. The magnitude errors with a 100 meV/ Å are indicated. **d**, Evolution of the Pt–C vertical distance during MD simulations of CO and H_2O adsorbed on the Pt(111) surface using the SNet-T25 model. Ivory, gray, white, and red spheres represent Pt, C, H, and O atoms, respectively. The moving-average curve is presented for SNet-T25; for DFT, the mean value is shown alongisde a horizontal dashed line; for UMA-OMat, -OC20, and -ODAC, only the values are provided. The term UMA-X is used, since the forward path of the UMA model is adapted depending on the task input X. Snapshots from the simulations with UMA-OMol and -OMC are also displayed, where the Pt surface structure was not preserved. **e**, As in **d**, except that CO is replaced by CO_2. The distance curves are displayed for the four MLIPs, whereas only the mean value is shown with a horizontal dashed line for DFT. The apparent truncation of the UMA curves arises from cases where the Pt–C distance exceeds 4.0 Å.

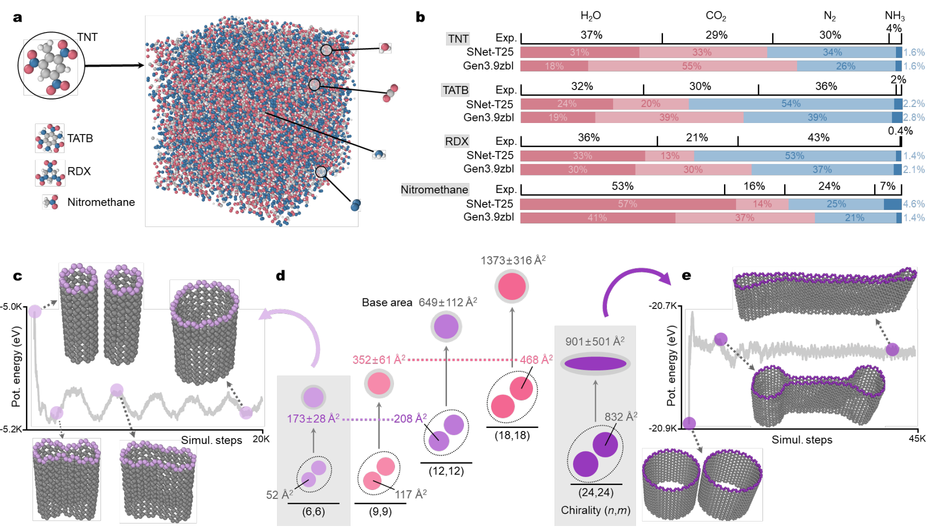

Fig. 4: **Atomistic simulations under extreme conditions: detonation products and CNT coalescence. a**, A snapshot of 1000 TNT molecules after a 40 GPa Hugoniostat simulation. Gray, white, blue, and red spheres represent C, H, N, and O atoms, respectively. For reference, single-molecule structures of TATB, RDX, and nitromethane are also shown; their post-simulation configurations are not displayed as they appear similar to TNT. **b**, Normalized product distributions after Hugoniostat simulations of each compound. We only consider H_2O, CO_2, N_2, and NH_3, as they constitute the main products of all four compounds. **c**, Potential energy evolution during the coalescence of two (6,6) CNTs. Representative snapshots of the initial, intermediate, and final states are included in the plot. Gray spheres indicate carbon atoms, while purple spheres denote the atoms used for area calculation. **d**, Schematic representation of CNTs as circles. Purple circles denote CNTs belonging to the (6,6) family, including (6,6) itself and those derived from it, whereas red circles denote CNTs belonging to the (9,9) family. The upper arrow represents coalescence, and the shading of the coalesced CNTs indicates the degree of projected area variation during the dynamic simulation. Coalescence of (24,24) CNTs forms an elliptical shape, indicative of their flatness. See Supplementary Fig. 10 for the variation of projected area with simulation step. **e**, Potential energy evolution during the coalescence of two (24,24) CNTs, plotted as in **c**.

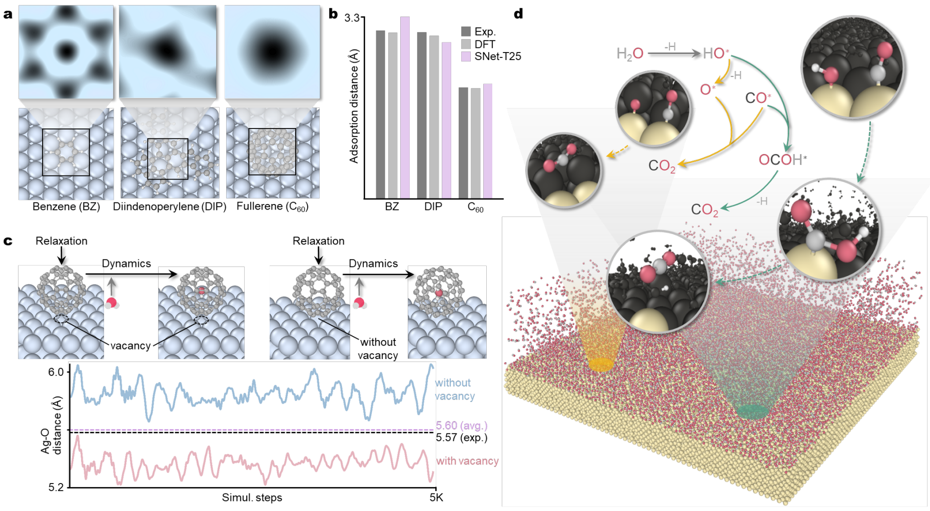

Fig. 5: **MLIP-MD simulations reproduce adsorption and catalytic reaction pathways. a**, Relaxed structures of π-conjugated molecules—BZ, DIP, and C_{60}—on the Ag(111) surface, together with their potential energy surfaces (PES) near equilibrium. Light-cyan, gray, and white spheres represent Ag, C, and H atoms, respectively. In the fullerene case, a vacancy is introduced at the center of the Ag(111) surface. Darker regions in the PES correspond to lower-energy configurations. **b**, Vertical distances of BZ, DIP, and C_{60} in their relaxed structures on the Ag(111) surface. The vertical distance is defined as the separation between the nearest carbon atom of each molecule and the Ag(111) surface plane. **c**, Evolution of the Ag–O distance during MD simulations. The red curve denotes the case of C_{60} encapsulating H_2O on Ag(111) with a central vacancy, whereas the blue curve denotes the defect-free Ag(111) surface. The averaged Ag–O distances of the two is denotes as a dashed line in pale purple (5.60 Å). (See Supplementary Fig. 12 for fixed-Ag(111) and C_{60} simulations.) **d**, Reaction mechanism of the water–gas shift obtained from MD simulations with H_2O and CO as reactants on a Au(100) surface terminated with OH groups. Ivory, gray, red, and white spheres represent Au, C, O, and H atoms, respectively. Black spheres highlight the atoms surrounding the reactive centers for visual emphasis.

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8 Methods

8.1 Data generator (DG)

The GAIA framework requires only minimal input data: Cartesian coordinates in xyz format for non-periodic molecules (NP) or a unit-cell description in POSCAR format for periodic solids (P). Either input can be provided independently, and both may be supplied. The input structures are then processed by the builders, as illustrated in Supplementary Fig. 1, to generate unlabeled configurations. These configurations are subsequently labeled with DFT energies and forces obtained from single-point calculations, ab initio molecular dynamics, and the initial step of the nudged elastic band method [59]. The subsequent subsections detail the algorithms used by each builder. These builders differ in how atomic species are arranged on the lattice or within the unit cell, thereby producing complementary sets of training configurations.

Checkerboard builder. The Checkerboard builder generates synthetic bulk-like structures by arranging two selected elemental species on a simple cubic lattice. The cubic cell has edge length L and is filled with atoms on a uniform grid of spacing s, resulting in $(L/s)^3$ atomic sites. For each element pair (A, B), stoichiometric ratios are sampled within a user-specified range $[r_{\min}, r_{\max}]$, including the symmetric 1:1 case. Atomic species are then distributed over the grid according to the chosen ratio, and their positions are randomly permuted to avoid artificial ordering. The cell is treated as periodic with cubic lattice vectors $(L, 0, 0)$, $(0, L, 0)$, and $(0, 0, L)$. The number of distinct stoichiometric ratios available for each pair is $q^{\mathrm{CB}} = k^2 - k + 1$, where $k = r_{\max} - r_{\min} + 1$. For example, $k = 1$ yields only the 1:1 ratio, while $k = 2$ includes 1:1, 1:2, and 2:1. Across all unordered element pairs, the total number of generated checkerboard structures is $q^{\mathrm{CB}} \cdot \binom{|\mathrm{e}|}{2} + |\mathrm{e}|$, where $|\mathrm{e}|$ is the cardinality of a set of the input elements, namely the number of available elements. Minor deviations of up to one atom from the exact stoichiometry can occur due to integer rounding.

Bulk builder. The Bulk builder generates bulk variants from P, of which the elements are expanded into a supercell where the replication factors (n_a, n_b, n_c) are chosen such that the total number of atoms does not exceed the predefined threshold value (e.g., 250). Among all valid candidates, the supercell with the largest atom count is selected. From the resulting supercell, a series of perturbed structures is produced. First, the cell volume is modified by scaling factors of 1.1 and 1.2 relative to the original size. In addition, a fraction of atoms is randomly removed, with deletion ratios of 0.1 and 0.2. For each atom-deleted configuration, additional perturbations are introduced by isotropically scaling the volume with factors of 0.85, 0.90, 0.95, and 1.05. This procedure generates 13 structures per P input: one unperturbed supercell, two volume-scaled cells, two atom-deleted cells, and eight atom-deleted cells with additional volume scaling. In total, $13|\mathrm{P}|$ configurations are generated and stored for subsequent DFT labeling.

Slab builder. The Slab builder shares the same supercell construction and perturbation procedures as the Bulk builder, but differs in converting the bulk supercell into a slab geometry. The bulk cell is cleaved along the (100) plane, and a vacuum gap of 30 Å is inserted along the third lattice vector $\mathbf{c}$ of the simulation cell. The structure is then translated so that the bottom surface lies at $z = 0$. To maintain structural stability, atoms are grouped into layers by their z-coordinates within a tolerance of 0.1 Å, and 20% of the lowest layers are constrained during subsequent calculations. This process ensures that the resulting slab retains in-plane periodicity while exposing a surface suitable for DFT labeling.

Adatom builder. The Adatom builder shares the same supercell construction procedure as the Slab builder, but differs by placing single-atom adsorbates on the slab to sample a coarse potential-energy surface (PES) above the exposed face. Adsorbates are defined from the unique set of elemental species present in the available P and NP inputs. For each element, a single atom is positioned on a two-dimensional grid covering half of the surface unit cell, with a 1 Å margin from the edges. The grid is discretized into r^{AD} points (5×5 by default). At each in-plane grid point, the adsorbate is placed at a series of discrete heights $d^{\mathrm{AD}} = 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70, 0.80, 1.00, 1.20, 1.40, 1.60, 2.00$ Å above the slab, relative to the highest surface atom. The sampling is denser near the surface and becomes progressively sparser at larger distances, providing finer resolution for short-range interactions and coarser resolution for the asymptotic region. By default, the number of configurations generated is $N^{\mathrm{AD}} = (\binom{|\mathrm{e}|}{2} + |\mathrm{eP}| - \binom{|\mathrm{eNP}|}{2}) \cdot |r^{\mathrm{AD}}| \cdot |d^{\mathrm{AD}}|$ where $|\mathrm{eP}|$ the number of P elements, and $|\mathrm{eNP}|$ the number of NP elements. The N^{AD} accounts for all possible element pairs, adds the self-pair correction for eP, and excludes pairs composed solely of eNP.

Admol builder. The Admol builder constructs adsorbed–molecule configurations on slab surfaces. Conceptually, it follows the same workflow as the Adatom builder, except that the adsorbates are NP rather than single atoms. The NP molecules serve as adsorption candidates and are subsequently aligned, replicated, and placed above the slab. For each P–NP pair, a surface supercell is prepared and the NP molecule is cell-synchronized to the surface reference so that its fractional extents fit within the surface cell. Integer repeat numbers (N_x, N_y) are chosen such that the in-plane dimensions of the molecular supercell match those of the surface cell to within 0.5 Å. A rotated molecular supercell is generated by tiling the molecule r^{AM} times, where $r^{\mathrm{AM}} = N_x \times N_y$; each tile is assigned a random three-dimensional rotation to diversify orientation and registry. The tiled molecule is recentred laterally over the surface and placed at discrete adsorption heights d^{AM} of 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, and 4.5 Å above the topmost surface atoms. This placement is repeated over $|r^{\mathrm{AM}}|$ (20 by default) independent trials to sample orientations and lateral registries. By default, the number of configurations generated is $N^{AM} = |\mathrm{NP}| \cdot |r^{\mathrm{AM}}| \cdot |d^{\mathrm{AM}}|$ per P input.

Nanoreactor$^+$ builder. This builder explores the reactive configurational space of the input molecular systems by combining the quantum cluster growth (QCG) scheme [60] with the metadynamics (MTD) simulations. This two-step procedure first generates candidate reactant assemblies and then drives them through reactive trajectories to sample bond breaking and formation events.

The QCG procedure was implemented using CREST [60, 61] in the QCG mode. Each atomic element contained in a P, when available, was first paired with a single NP to form a P–NP combination. These pairs were then extended by combining the P–NP assemblies with additional NP partners. In the absence of P, one NP was fixed as the solute and combined with another NP. For each pair, the number of partner molecules was controlled by an adjustable parameter. Spin multiplicities were initially set to closed-shell configurations. An initial geometry optimization could be skipped when the input structures were already equilibrated. A short (1 ps by default) MD ensemble was performed in the ALPB water model, while an outer wall potential confined the cluster with a scaling factor of 1.2 by default. The lowest-energy structures were collected and passed to subsequent Nanoreactor^{+} simulations. If convergence failed, the workflow automatically retried with progressively altered parameters: the wall scaling factor was varied sequentially (1.2, 1.4, 0.8, 1.0, 1.6), and the spin multiplicity was cycled through singlet, doublet, and triplet states. If all combinations failed to yield a converged ensemble, the corresponding pair was discarded as a failed QCG attempt. By default, the ideal number of QCG generated configurations is $N^{\mathrm{QCG}} = |\mathrm{P}| \cdot |\mathrm{NP}| \cdot (|\mathrm{NP}| - 1) + |\mathrm{NP}|^2$.

After the QCG, the builder explores reactive events of the QCG-generated configurations by performing MTD simulations with the xTB engine [62]. For each input structure, a confining sphere was estimated from a short CREST calculation; when this failed, a random radius between 6 and 10 Å was assigned instead. The working radius for each trial was then drawn uniformly between this value and 1.5 times it, ensuring variability across runs. At each trial, the log-Fermi wall potential (6000 K) and biasing parameters (k_{push}, α, and β) were re-sampled randomly within predefined ranges. Spin multiplicities were tested sequentially in a fallback manner: the first attempt used the doublet state, followed by singlet, triplet, and quartet. If the quartet also failed to converge, the corresponding MTD simulation was discarded. Simulations were run for 10 ps with a 1 fs integration step, without SHAKE constraints. The MTD bias was updated every 10 fs, and trajectories were saved every 50 fs, producing 200 snapshots per run. Each trajectory was postprocessed by converting individual snapshots from XYZ to SDF format with Open Babel [63], and the converted structures were parsed with RDKit [64] to construct bond-order heatmaps, the bondmap. The bondmaps were averaged across all snapshots within a trajectory to obtain a representative bond-order matrix. Newly generated matrices were systematically compared with the entire set of previously stored matrices using the structural similarity index (SSIM), and only those below a similarity threshold were retained as distinct products. As a result, the total number of configurations generated by the Nanoreactor^{+} builder is $N^{\mathrm{NANO}} = k^{\mathrm{NANO}} \cdot s^{\mathrm{NANO}}_{\mathrm{MTD}} \cdot N^{\mathrm{QCG}}$ where k^{NANO} is the user-defined number of unique bond-order matrices retained per configuration, $s^{\mathrm{NANO}}_{\mathrm{MTD}}$ is the number of snapshots generated in each MTD trajectory, and N^{QCG} is the number of input structures obtained from the QCG stage. See Supplementary Fig. 2 for representative configurations of Nanoreactor^{+} builder.

Supercell with random rotation. The resulting Nanoreactor^{+} configurations were combined into supercells before DFT labeling. Replication factors along the three lattice vectors were chosen to maximize the number of atoms while keeping it below a user-defined limit per snapshot. To avoid generating identical repeats, each replica was rotated as a rigid body using a rotation matrix defined by a random unit axis and a rotation angle uniformly sampled between 0 and 360°. Atomic coordinates were shifted to the center of mass, rotated, and then translated to the corresponding lattice position. The final simulation cell was defined with a small padding of 0.3 Å along each lattice direction.

Diatomic potential energy curve. Potential energy curves were generated for each unique pair of elements, with interatomic distances sampled uniformly from the factor of user-defined minimum to maximum value to the sum of their covalent radii (for Titan25, 0.6 and 3.0 values are used). The resulting energies and forces were then discretized into 30 bins and smoothed using cubic spline interpolation.

Dataset visualization. Fig. 2b was generated using the uniform manifold approximation and projection [65]. For each data point in the Titan25(G), embeddings were obtained from feature vectors extracted by the model trained on Titan25(G+I). Each feature vector corresponds to the aggregated output of the final representation layer, computed from the atomic-level embeddings of the input structure. Further details on the model architecture and training are provided in Method 8.4.

8.2 DFT labeling for Titan25

All DFT calculations were performed with VASP.6.4.1 [66, 67] Projector-augmented wave (PAW) pseudopotentials and the PBE functional were used. A plane-wave cutoff energy of 520 eV was applied, and van der Waals interactions were included through the D3 correction with zero-damping function. Brillouin-zone sampling was restricted to the Γ point. Electronic convergence was set to 10^{-5} eV, and atomic forces were converged to below 10^{-4} eV/ Å. Partial occupancies were treated using Gaussian smearing with a width of 0.05 eV. Symmetry constraints were turned off. Configurations generated by the Adatom and Admol builders were evaluated by single-point energy calculations under these conditions.

Structure relaxation. For the configurations generated by the Checkerboard builder, structural relaxations were performed using the conjugate-gradient algorithm with a relaxation step size of 0.2 for up to 20 ionic steps. During these relaxations, only the atomic positions were optimized while the cell shape and volume were kept fixed. Whether converged or not, the resulting structures were used as initial conditions for subsequent ab initio molecular dynamics (AIMD) simulations. These relaxations served only to generate suitable starting configurations for AIMD, rather than to obtain fully optimized structures.

Ab initio MD. Two AIMD protocols were used to generate DFT-labeled datasets, with each protocol applied to configurations generated by different builders. The first protocol was carried out in the NVT ensemble using the Nose-Hoover thermostat for 1,000 steps with a time step of 0.5 fs, during which the temperature was increased from 400 K to 600 K. This protocol was applied to datasets generated by the Checkerboard, Bulk, and Slab builders. The second protocol employed the same simulation parameters as the first, but the trajectories were limited to 20 steps. This shorter protocol was applied to Nanoreactor^{+} configurations. In both protocols, a fictitious Nose-Hoover mass parameter of 20 was used.

NEB initial step. Trajectories generated by the Nanoreactor$^+$ builder were further used to construct connected structures via the nudged elastic band scheme. Representative snapshots were taken at regular intervals, and initial and final configurations were paired to produce intermediate images through linear interpolation using the `nebmake.pl` utility provided with VASP. These interpolated configurations were not subjected to full NEB optimizations; instead, they were directly evaluated by single-point DFT calculations to provide reference energies and forces. Before the DFT-labeling, all generated structures were screened for unphysical short contacts. A Coulomb-like repulsion estimate was used, defined as $E_{\mathrm{rep}} = k/r$, where $k = 14.4$ eV $\cdot$ Å and r is the shortest interatomic distance. Configurations with $E_{\mathrm{rep}} > 50$ eV were discarded, and only the remaining physically reasonable structures were retained. As described in Method 8.1, the retained configurations were further expanded into supercells with randomized rigid-body rotations to enhance structural diversity.

8.3 Data improver (DI)

As outlined in Fig. 2c, DI serves to analyze the deficiencies in the exploratory data from DG and to supplement them accordingly. Its detailed procedure is described as follows.

Categorical analysis. As the first stage of DI, categorical analysis is performed on the generated data points to explore undersampled areas in the chemical space, which can be low-density regions or error-prone domains. The key here lies in establishing criteria for data categorization; we propose a two-dimensional scheme for it.

In the scheme, one axis corresponds to the conventional measure of total energy, the other to a diversity metric d_Λ, given by the standard deviation of eigenvalues of interatomic distances in a snapshot. Specifically it is obtained as $d_\Lambda(i) = (\sum_{j=1}^{n_i}(\lambda_j - \bar{\lambda})^2/n_i)^{1/2}$, where n_i is the number of atoms in snapshot i, and λ_j is the j^{th} eigenvalue of the atom-atom distance matrix. The proposed metric thus indicates how much configurational diversity is presented in the snapshot.

We note that variations in the number of atoms across data points were not explicitly handled in the measurement of the standard deviation. This could introduce biases in purely geometric comparisons, since the corresponding normalization was not applied. Nevertheless, such normalization could, in turn, suppress genuine contributions from supercell enlargement, such as inter-complex interactions with random orientations, which are essential sources of structural diversity. Hence, the design choice of omitting normalization reflects our focus on capturing not only shape-based differences but also the overall interaction modes relevant for MLIP training. Yet, to mitigate the possible geometric mismatch arising from variations in atom count, the maximum system size in a data point was restricted to 250 atoms. With this perspective, leaving the raw spectra allows accounting for the comprehensive modes in atomic environments represented across snapshots, thereby enabling the remaining stages to cover as broad a structural variety as possible.

With the user-defined values M_E and M_Λ, the two-dimensional space is divided into a $M_E \times M_\Lambda$ grid, where each cell of this grid has a value of the number of data points in the training dataset and the ratio of outliers in the validation dataset. Outliers refer to data points with prediction errors exceeding the user-defined threshold. In our case, we used force RMSE as the error metric and set the threshold to 2.0. Accordingly, cells with lower values for the training dataset and higher values for the validation dataset suggest the presence of sparse (or absent) coverage across both observable and hidden regions.

Basis snapshots sampling. Guided by the improvable coverage above, basis snapshots are data points, from which the new seed structures are selected to further span the current chemical space. To do this, we construct the probability distributions for the sampling, defined by $P_F(x_{ij}) = (\sum_{i',j'} x_{ij}/x_{i'j'})^{-1}$ and $P_P(\xi_{ij}) = \xi_{ij}/\sum_{i',j'} \xi_{i'j'}$ for the frequency-based sampling on the training dataset and the prediction-based sampling on the validation dataset, respectively, where x_{ij} is the number of data points in the cell C_{ij}, and ξ_{ij} is the ratio of outliers therein. In the current implementation, the number of basis snapshots is user-specified for both sampling strategies.

Seed expansion. Serving as the core of the seed expansion, the extraction of chemical substructures is performed on the selected basis snapshots. First, the positional coordinates of a snapshot are unwrapped in order to restore continuity across periodic boundaries. By creating an imaginary cube with side length $2\delta r_{\mathrm{COV}}$ around each atom, where r_{COV} is the covalent radius and δ is the volume tolerance, bond presences are identified by the overlap of the cube pairs. Accordingly, chemical substructures in each snapshot are then extracted by the depth-first search. Among them, those with less than a specific number of atoms—the minimum value within the Nanoreactor$^+$ results from the previous DG module—are filtered out. Finally, new seeds are determined from the extracted substructures in consideration of the previous N_{NANO}, which are fed back into the Nanoreactor$^+$ to further generate the exploratory data points.

8.4 Model training

To train MLIPs on Titan25 and other datasets for comparison, we used an equivariant graph neural network—specifically SevenNet [68], a variant of NequIP [69] in which the self-connection parameters are shared across all chemical elements. The model architecture was configured with an embedding block having a 128×0e output dimension and three interaction blocks, where the first two have output features of 128×0e+64×1e+32×2e+32×3e and the last has 128×0e, followed by the readout block. The training was performed using an in-house framework built on top of FAIRChem [70], extended to support the SevenNet and to enable the fused kernels such as cuEquivariance [71] and FlashTP [72]. For a fair comparison across datasets, the number of training steps was fixed at approximately 0.5 million with the same batch size. Accordingly, depending on dataset size, the model was trained for 100, 125, and 80 epochs on MPTrj, Titan25(G) and Titan25(G+I), respectively. In the case of ANI-1xnr, which has a much smaller dataset size than the others, there could be a risk of overfitting with overly prolonged training; thus, the training was limited to 300 epochs. During dataset preparation, since the test datasets in GAIA-Bench were constructed using PBE-D3, ANI-1xnr was re-labeled with the PBE-D3 functional under the same conditions as in Method 8.2. For MPTrj, D3 correction terms were calculated independently and added for a consistent comparison. The data in each dataset were split into training, validation, and test sets with an 8:1:1 ratio. Further training details are provided in Supplementary Table 1; the validation results on Titan25(G+I) are included in Supplementary Figs. 4 and 5.

8.5 GAIA-Bench

GAIA-Bench comprises four benchmark tasks: mol2mol, bulk, slab, and mol2surf, with reference data obtained from DFT calculations. The same DFT parameters and conditions as in Titan25 were applied. In the following section, we elaborate on the implementation details of GAIA-Bench.

Structure perturbation. For force evaluations, perturbed configurations were generated by applying randomized displacements to the same structures used for energy comparisons in each benchmark. Each perturbed structure was obtained through multiple trial displacements, with only those meeting the acceptance criteria retained. At each trial, atoms were randomly displaced by 0.01–0.20 Å (0.01–0.1 Å for mol2mol) along directions biased by local neighbor vectors, with small random angular deviations added to the displacement directions. If the target force distribution was not sufficiently sampled after 10 iterations, and the maximum atomic force remained below 1.2 times the target, the upper displacement bound was gradually increased by 1% per step until a maximum of 1.0 Å was reached. Perturbed structures were accepted only if at least 30% (20% for mol2mol) of atoms had forces within the target range, while the maximum force remained below 1.2 times the target. For mol2mol, target forces of 0.1, 1.0, 3.0, and 5.0 eV/ Å were considered, whereas bulk and slab employed targets of 3.0, 5.0, and 7.0 eV/ Å . A total of 3,024, 4,713, 870, and 80,424 configurations were generated for force calculation in mol2mol, bulk, surf and mol2surf, each. The axis-wise force MAE values were then evaluated against the DFT forces, with the analysis performed for each group as defined within the individual benchmark tasks.

mol2mol. The molecular interactions of a set of nine molecules were investigated: 1,2,4-trimethylbenzene, 1,2-ethanediol, 1-butanol, 1-propanol, 2-butanol, 1,4-dioxane, nitroethane, propylamine, and water, consisting solely of C, H, N, and O. Structural information for these molecules was retrieved from PubChem [73]. The near-equilibrium structures of molecular pairs, including self-pairs, were identified using the QCG scheme with the GFN2-xTB method [60]. From those near-equilibrium structures, one molecule was rotated around a random axis to generate 30 distinct orientations, sampling diverse molecular interactions. From each rotated molecular pair, the interatomic vector connecting the closest atoms between the two molecules was determined and used as the reference axis. Along this axis, the intermolecular separation was scaled from 0.5 to 3.5 times the original distance. The resulting 22,650 configurations were classified into 81 distinct groups based on the molecular components of the pairs. For each group, the relative energies were evaluated with respect to the configuration having the largest intermolecular separation, enabling the calculation of interaction energies.

bulk. Periodic bulk structures were obtained from the Materials Project [74], considering assemblies of Ag, Au, Cu, Pd, Pt, Rh, Ru, O, and H. All unary, binary, and ternary combinations were included, except for sets containing only O and H. Supercells were constructed by repeating the conventional cell to maximize the number of atoms under a cap of 200. To sample volumetric strain states, the cells were isotropically scaled with factors of 0.85, 0.90, 0.94, 0.98, 1.00, 1.02, 1.04, 1.06, 1.08, 1.10, 1.12, 1.14, and 1.16, with atomic positions scaled accordingly, yielding a family of uniformly strained supercells for each composition. In total, 1,783 configurations were generated by this procedure.

To construct energy–volume curves, structures sharing the same composition were grouped. DFT calculations that did not converge or failed were removed, which sometimes reduced the group size. Only groups retaining at least two valid structures were used in the analysis, resulting in 143 valid groups out of 144. For each group, the structure with the lowest DFT total energy was identified and designated as the reference geometry. This geometry was then re-evaluated with the MLIP, and relative energies were computed by subtracting the corresponding reference energy in each method. This procedure ensured a consistent baseline while allowing direct comparison between DFT and a MLIP.

slab. The structures for the slab benchmark were obtained similarly to the bulk benchmark, but only binary and ternary combinations were considered. Surface indices (001), (110), and (111) were generated with two layers and a vacuum spacing of 30 Å, ensuring the total number of atoms did not exceed 400. A total of 392 configurations were generated for the slab benchmark.

With reference to the (110) surface, the relative energy of (001) and (111) surface for the same components are considered as a group. If the DFT calculation for the (110) surface fails for a group, the (111) surface is used instead. A total of 167 groups are categorized.

mol2surf. The mol2surf benchmark comprises configurations of molecular bilayers on periodic surfaces. These surface structures are generated using the same methodology as the slab benchmark, with molecules taken from the mol2mol benchmark set. In contrast to the slab benchmark, the mol2surf benchmark considers only the (001) and (111) surface indices, with a maximum of 200 atoms in the supercell. For each surface, two molecular pairs are chosen and arranged in a bilayer configuration with a 1.5 Å offset. The molecules are then replicated with random rotations, following the DG procedure. For each configuration, three random rotations of the bilayer molecules are performed, and the resulting configurations are grouped together. A total of 80,424 configurations were generated, resulting in 26,908 groups. The energy of each group was assessed relative to its first configuration.

8.6 MLIP-MD simulations

MD simulations were conducted mainly using LAMMPS version 2 Aug 2023 [75], employing the model trained on Titan25, SNet-T25, with its checkpoint converted for parallel simulations.

Pt-COx-H2O. To investigate CO or CO_2 on the Pt(111) surface in an explicitly solvated environment, MD simulations were performed. The initial cubic Pt unit cell structure was taken from the Materials Project (ID: mp-126). Supercells with fewer than 500 atoms were constructed to create four-layer Pt(111) surfaces for both CO and CO_2 cases. For CO, 96 molecules formed a double layer, overlaid by 256 water molecules in four layers. For CO_2, 36 molecules were placed on the surface in a double layer, followed by 125 water molecules in five layers.

Both simulations were conducted using the Nose-Hoover thermostat at 300 K. The simulations consisted of 50,000 steps with a 0.5 fs timestep, resulting in a total simulation time of 25 ps. Snapshots were recorded every 100 steps to analyze the vertical distance between the plane constructed from the atoms in the outermost Pt surface layer and the C atoms. For each snapshot, we computed the perpendicular distance between this reference plane and all carbon atoms. A global cutoff (1.5 Å for CO and 4.0 Å for CO_2) was applied, such that only carbon atoms located within those distance from the Pt plane were included in the analysis. The reported values correspond to the mean perpendicular distance of these atoms at each snapshot, and trajectories in which no carbon atom satisfied the cutoff were excluded from the averaging procedure.

For both systems, simplified structures were used for additional DFT calculations. The four-layer Pt(111) surface, comprising 96 atoms, was used to place 6 CO (or CO_2) molecules in a double layer, accompanied by 24 water molecules in four layers. The simulations employed the PBE-D3 functional and Nose-Hoover thermostat at 300 K, consisting of 20,000 steps with a 1.0 fs timestep, resulting in a total 20 ps simulation.

By means of these simulations, the Titan25-trained model was compared against UMA-M [76]. The UMA-M model comprises approximately 50 million active parameters—nearly ten times larger than the total number of parameters in SNet-T25 (0.5 million parameters)—and was trained on a combined dataset spanning OC20 [77], OMat24 [78], OMol25 [79], ODAC25 [80], and OMC25 [81]. Even though its model parameters are shared across the five datasets, task-specific adjustments are applied during inference. We additionally note that, since the official UMA checkpoints [82] are incompatible with LAMMPS, the simulations were performed using ASE v3.25.0 [83] instead.

Hugoniostat simulation of energetic molecules. The isotropic Hugoniostat simulations were conducted on four energetic molecules: RDX, TNT, TATB, and nitromethane, whose molecular structures were from PubChem. For each system, one thousand molecules were placed in a cubic cell with a 4 Å offset applied to each molecule.

To identify products, molecules in MD trajectory files were analyzed using a graph-based approach where atoms were nodes and bonds were edges based on interatomic distances. Bonding was determined by checking if the distance between two atoms was within 0.3 to 1.5 times the sum of their covalent radii. Each resulting connected component represented a distinct molecular species. A molecule is identified as H_2O if the component consists of one oxygen and two hydrogen atoms and H–O–H angle falls within the range of 73–135.5° (0.7–1.3 × 104.5°). For CO_2, a component is counted as CO_2 if it consists of one carbon and two oxygen atoms and O–C–O angle is within 126–180° (0.7–1.0 × 180°). A molecular fragment was classified as NH_3 if it contained one nitrogen and three hydrogens, and all H–N–H bond angles lay within 30 % of the ideal tetrahedral angle (107°). For each species, the normalized molecular counts were averaged over the last ten snapshots of the trajectory.

CNT simulation. CNT structures were generated using TubeASP [84], with an initial C–C bond length of 1.42 Å. For each chiral index, the corresponding structure consisted of 12 unit cells. Initial configurations were prepared for pairs of CNTs with identical chiral indices, and the inter-CNT distance was set to 3.0 Å.

The coalescence of CNTs was simulated using a hybrid approach combining kinetic Monte Carlo (kMC) and MD methods, as implemented in the Papreca package [85]. At the contact interface between the two CNTs, the simulation protocol involved severing the C–C bonds at the interface between the two CNTs and subsequently performing 100 steps of NVT MD simulations at 1000 K with a timestep of 0.5 fs, without any additional bias on the chemical bonds. Since the rate constants were not derived from physical kinetics, the simulated time should not be interpreted quantitatively.

To analyze the cross-sectional geometry of carbon nanotubes, the atomic coordinates were projected onto a plane normal to the tube axis. DBSCAN clustering [86] was applied to the atomic positions in each frame to identify individual tube cross-sections. Ellipse fitting using PCA was performed on the projected coordinates to determine the semi-major and semi-minor axes. The cross-sectional area was then calculated using πab, where a and b denote the length of fitted semi-axes.

π-conjugated molecule on metallic surface. The unit cell structures for face-centered cubic metals (Ag, Au, Cu, Pt) were retrieved from the Materials Project. Using these unit cells, four-layer (111) surfaces with 256 atoms were constructed with a 30 Å vacuum layer. The molecular structures of benzene and diindenoperylene were retrieved from PubChem, and that of fullerene from ref. [87]. For the initial relaxation structure, the adsorbate was positioned 2.0 Å above the surface. For the fullerene system, a modified Ag(111) surface with a central vacancy was employed to replicate the experimental conditions. For H_2O encapsulated in the fullerene, the molecule was centered within the fullerene cage and subsequently placed on the Ag(111) surface, with configurations both with and without a central vacancy considered.

For both SNet-T25 and UMA-M models, relaxations were carried out using the BFGS algorithm in ASE, with atomic positions relaxed until the maximum residual force fell below 0.001 eV/ Å. MD simulations were conducted to determine the vertical distances between the Ag(111) surface (with and without vacancy) and the O atom of H_2O encapsulated within fullerene. The simulations employed the Nose-Hoover thermostat at 180 K and consisted of 50,000 steps with a 0.5 fs timestep, yielding a total simulation time of 25 ps.

MD simulation of the water-gas-shift reaction. The unit cell structure of Au was retrieved from the Materials Project (ID: mp-81). A five-layer Au(100) surface was then generated by replicating the unit cell, yielding a total of 69,620 Au atoms. Random termination with 4,462 OH molecules on the Au surface was also performed. After OH termination, 4,332 CO molecules were deposited in three layers (1,444 molecules per layer), followed by three additional layers of H_2O comprising 4,557 molecules in total (1,519 per layer), with a 30 Å vacuum layer.

MD simulations were performed using the Nose-Hoover thermostat at a temperature of 600 K. The bottom layer of Au atoms was fixed, and a reflecting wall was placed at the 80 Å from the bottom of the simulation box to confine the system along the z-direction. The simulation consisted of 200,000 steps with a 0.5 fs timestep.

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9 Supplementary information

9.1 Related work

In this subsection, we review prior work and highlight the contributions of GAIA.

Datasets. Since the Titan25 dataset has been developed for general-purpose use, here we focus primarily on the corresponding datasets, rather than system or task-specific ones.

Following the advent of Materials Project [1], numerous datasets have been introduced for bulk crystal structures. MPF [2] and MPTrj [3] were pioneering efforts that curate data from MP into MLIP datasets at the million scale. Subsequently, the dataset size has progressively grown, through GNoME [4], MatterSim [5], and Alexandria [6], culminating in the OMat24 dataset [7] containing 100 million snapshots.

Moving from bulk structures, OC20 [8] and OC22 [9] addressed the adsorption of CHNO-based adsorbates on slabs, thereby capturing interfacial interactions at surfaces, which are also among the central goals of our approach. ODAC23 [10] and ODAC25 [11] focused on a different type of adsorption, namely the adsorption of CO_2, H_2O, N_2, and O_2 on metal-organic frameworks. In addition, the molecular datasets such as OMol25 [12] and OMC25 [13] were developed very recently. OMol25 provides extensive coverage of intra- and intermolecular interactions, including diverse charge and spin stages, solvation, and reactivity. OMC25 was introduced as a molecular crystal dataset.

In the context of chemical reactivity, the Transition1x [14] dataset incorporated reactivity by leveraging the nudged elastic band [15] to obtain transition states connecting reactants and products. The OMol25 dataset has sought to achieve this through geodesic interpolation [16] and sampling based on the artificial force induced reaction [17], which enables the identification of numerous reaction pathways by applying artificial forces to specific atom pairs (or fragments) to induce bond rearrangements. However, the targeted nature in searching constrains these approaches from performing comprehensive global exploration. To this end, Nanoreactor [18] offers a promising alternative, providing efficient discovery of reactive events with minimal bias under high-temperature conditions. Using the Nanoreactor, ANI-1xnr [19] was developed for four elements, C, H, N, and O.

Beyond the prior efforts, the proposed GAIA and the resulting dataset Titan25 is characterized by three distinctive features:

- GAIA facilitates de novo design of exploratory structures, which, in principle, allows for an unbounded number of varied snapshots to be generated. By contrast, most of the existing datasets fully or partially rely on reference databases for structure construction. For instance, the crystalline datasets and the bulk structures in OC20 and OC22 were based on the Materials Project database, the metal-organic framework datasets on CoRE MOF 2019 [20], OMC25 on the OE62 dataset [21], and OMol25 on multiple databases.
- GAIA can provide effective diversity to represent a wide range of chemical reactions. The diversity is established based on Nanoreactor, as implemented in ref. [22], where the expensive ab initio treatment is replaced by the semi-empirical tight-binding

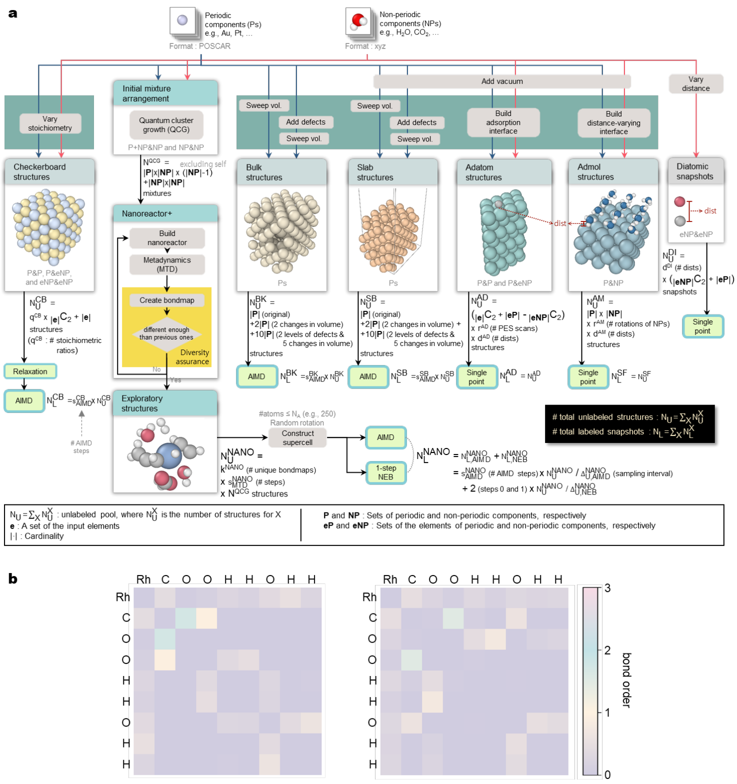
method with metadynamics. The implementation can provide higher efficiencies in capturing rare events compared to ANI-1xnr. This is because, in building the ANI-1xnr dataset, the Nanoreactor was powered by molecular dynamics with temperature and density oscillations, which, albeit driven by an MLIP, still inherently involved considerable computational cost. In addition, whereas the MLIP-driven molecular dynamics for ANI-1xnr requires a pre-existing (seed) dataset to train the corresponding MLIPs, our approach is capable of generating structures from scratch. Furthermore, as an extension of Nanoreactor, we introduced three additional treatments to Nanoreactor$^+$ to achieve better exploration: (i) Quantum cluster growth [23] is adopted for the initial mixture arrangement of the user-specified input components, rather than relying on manual or random placement. The resulting arrangements serves as the initial configuration for the exploration. (ii) The outputs of Nanoreactor are filtered according to the bondmap similarity criterion, discarding results similar to previously generated ones and collecting until the desired number is reached. (iii) The random rotation is applied during the repeated placement of the generated structures to construct supercells, thereby better capturing the interactions involved.

- GAIA outperforms the state-of-the-art datasets on interfacial reactions. In a direct sense, this is accomplished by the five determined structure builders, while diverse interactions of metals or metalloids with nonmetallic components stem from the exploratory builder, Nanoreactor$^+$. In our evaluation, the Titan25-trained model achieved the improved results on the mol2surf task in the GAIA-Bench compared to the ones trained on MPTrj and ANI-1xnr (Fig. 3a,b,c). It also maintained its superiority under the actual simulations, outperforming the UMA model [24] trained on multiple foundation datasets in a unified manner (Fig. 3d,e).

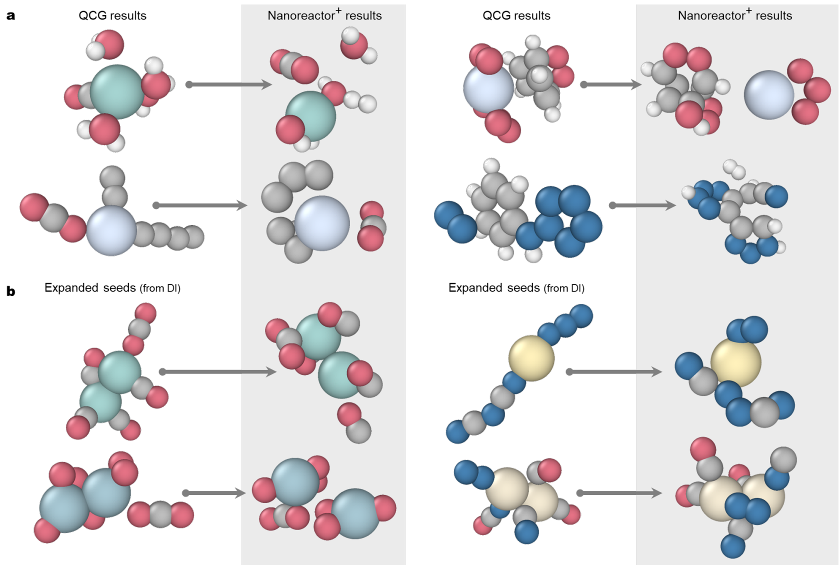
ML-guided structure augmentation. In one line of research, there have been active learning studies that leverage the concept of uncertainty for the structure augmentation. The uncertainty here is usually defined as the degree of disagreement among an ensemble of models (e.g., the query-by-committee [25]), typically quantified by the variance of their predictions. In UDD-AL [26] and HAL [27], the uncertainty estimation is applied in two aspects: (i) it operates as a bias in the dynamics so that the sampling is performed toward regions of high uncertainty, and (ii) structures exhibiting high uncertainty during the dynamics are prioritized for quantum-mechanical labeling, thereby promoting an efficient expansion of the accessible chemical space. There also exist approaches that employ static optimization methods, such as NEB and SE-GSM [28], to explore reaction pathways, as an alternative to the dynamics-based strategies [29, 30]. In this regard, a recent work, ArcaNN [31], provides a structural framework for the automated implementation of such uncertainty-based methodologies.

However, these methods are constrained in two ways: First, the dynamics for structure search starts from snapshots in a candidate reservoir—such as an a priori structure pool or its augmented versions through active learning iterations—serving as initial configurations. On the other hand, in the structure augmentation stage of GAIA, Nanoreactor$^+$ operates on new substructures extracted from existing data points produced by the data-generator, yet these are fundamentally distinct from the seed components previously provided by the user. Second, the active learning methods incur a non-negligible amount of additional cost in training multiple models to build an ensemble. That is, if the ensemble consists of N (e.g. 8) MLIPs, each of them needs to be trained separately. Although several studies employed a concept of evidential learning [32, 33] to estimate the uncertainty with only a single model [34, 35], the burden of model training still remains. By contrast, the data-improver of GAIA does not require the substantial overhead beyond the structure search process; it only demands the model inference cost for the prediction-based sampling and the extraction cost of chemical substructures, far lower than that of model training. In this context, several works have attempted to avoid reliance on pre-existing models in structure search [36, 37], akin to ours, but their basic strategy originates from random sampling, which may lead to inefficiency in exploring chemical space.

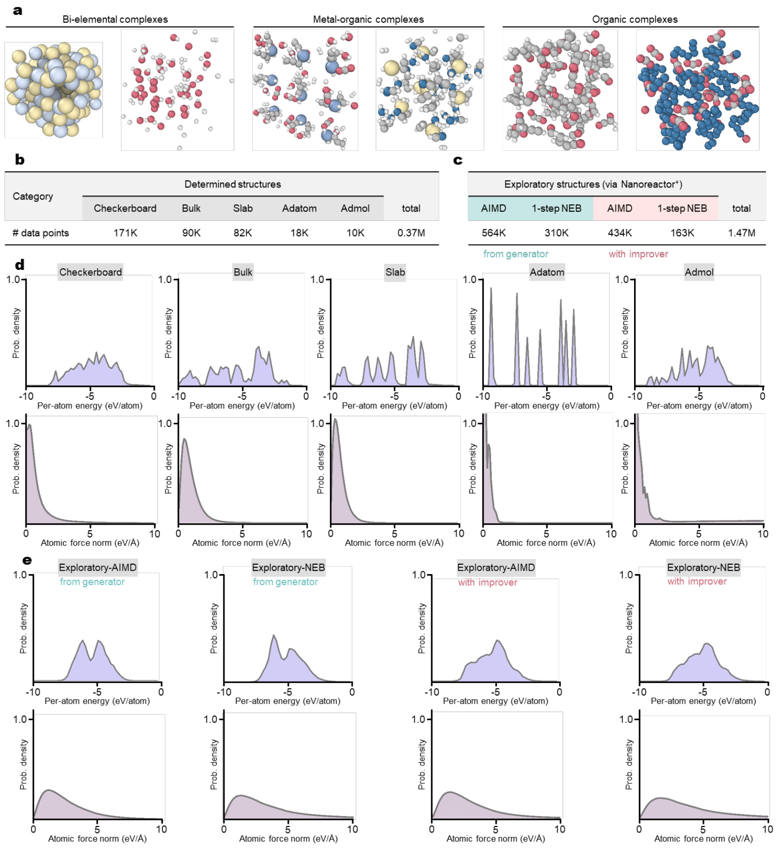
9.2 GAIA and Titan25



Supplementary Fig. 1: **a**, Overview of GAIA data-generator. Various combinations of atomic arrangements are generated using the user-provided inputs (top) and the builders described below. Representative cases of structures constructed by each builder are illustrated. $N_{\mathrm{U}}^{\mathrm{CB}}$, $N_{\mathrm{U}}^{\mathrm{NANO}}$, $N_{\mathrm{U}}^{\mathrm{BK}}$, $N_{\mathrm{U}}^{\mathrm{SB}}$, $N_{\mathrm{U}}^{\mathrm{AD}}$, and $N_{\mathrm{U}}^{\mathrm{AM}}$ indicate the maximum number of unlabled structures that can be generated by each builder (Checkerboard, Nanoreactor^{+}, Bulk, Slab, Adatom and Admol) and $N_{\mathrm{U}}^{\mathrm{DI}}$ for diatomic curve add-on. The same symbols with the subscript "L" denote the maximum number of labeled structures. For detailed explanation, see Method. **b**, Illustration of bondmap. Averaged bond matrices of trajectories generated from a mixed structure comprising a Rh single atom, a single CO molecule, and two H_2O molecules are shown. The left and right panels depict the first and second unique averaged bond matrices derived from this mixed structure, obtained with an SSIM threshold of 0.98.



Supplementary Fig. 2: **a**, Four representative transformations from input (QCG results) to output (Nanoreactor$^+$ results), indicated by arrows, during the data-generator stage. Spheres are color-coded by element; larger spheres represent metals, whereas smaller spheres represent organic elements. **b**, As in **a**, but during the data-improver stage. The inputs are expanded seeds, and the outputs are again the Nanoreactor$^+$ results.



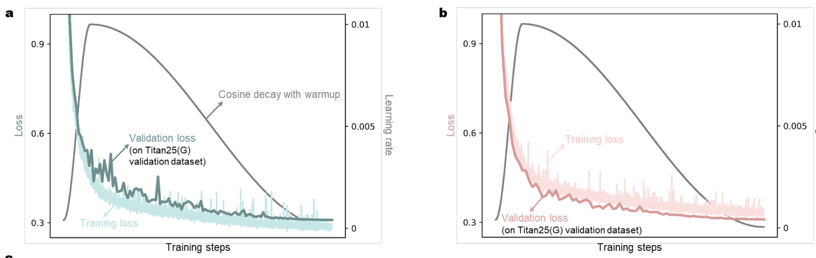
Supplementary Fig. 3: **a**, Representative configurations in the Titan25 dataset: bi-elemental complexes (Checkerboard builder), and metal–organic and organic complexes (Nanoreactor$^+$ builder). **b**, Number of labeled configurations in the Titan25 dataset for each builder (excluding the Nanoreactor$^+$), collectively termed determined structures. **c**, Total number of labeled configurations generated from Nanoreactor$^+$, including those from the data-generator and the data-improver, designated as exploratory structures. **d**, Probability densities of per-atom energy (upper) and atomic force norm (lower) of determined structures, grouped into five pairs. Builder names are annotated at the center of each plot. **e**, As in **d**, but with exploratory structures generated by the data-generator (left two pairs) and the data-improver (right two pairs). Results labeled by AIMD and NEB are shown separately.

9.3 Model training

Supplementary Table 1: Hyperparameter details

Irreps of embedding blocks	128×0e			
Number of interaction blocks	3			
Irreps of interaction blocks	128×0e+64×1e+32×2e+32×3e 128×0e+64×1e+32×2e+32×3e 128×0e			
Cutoff radius (Å)	6			
Cutoff function	XPLOR [38]			
Radial basis function	Bessel			
Number of radial basis functions	8			
Energy scaled by	Root-mean-square of forces			
Energy shifted by	Per-atom energy mean			
Optimizer	AdamW			
Batch size	256			
Learning rate schedule	Cosine			
Warmup epochs	10% of training			
Maximum learning rate	0.01			
Gradient clipping	100			
Weight decay	0.001			
Energy loss (E)	Per-atom MAE			
Force loss (F)	L2MAE [39]			
Stress loss (S)	MAE			
Loss coefficients	E:F:S=1:1:1			
Datasets	ANI-1xnr	MPTrj	Titan25(G)	Titan25(G+I)
Training steps	167K	493K	486K	498K
Training epochs	300	100	125	80

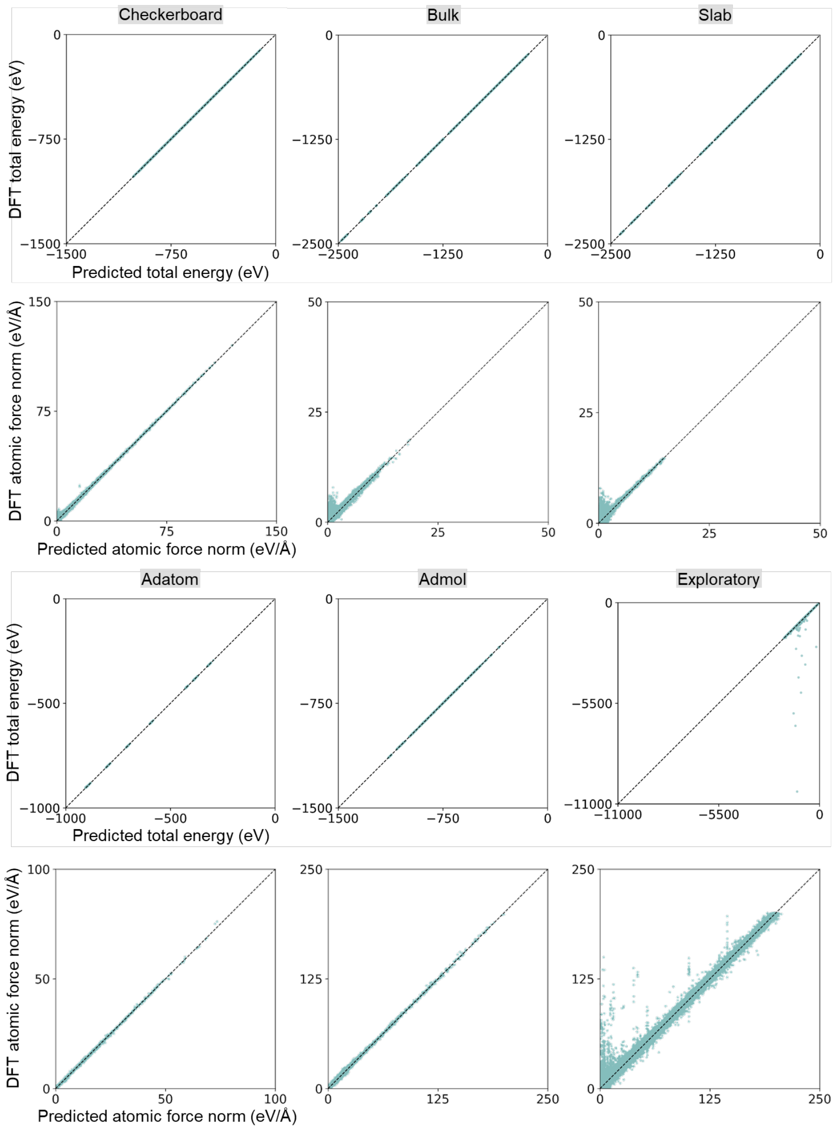
9.4 Validation performance on Titan25



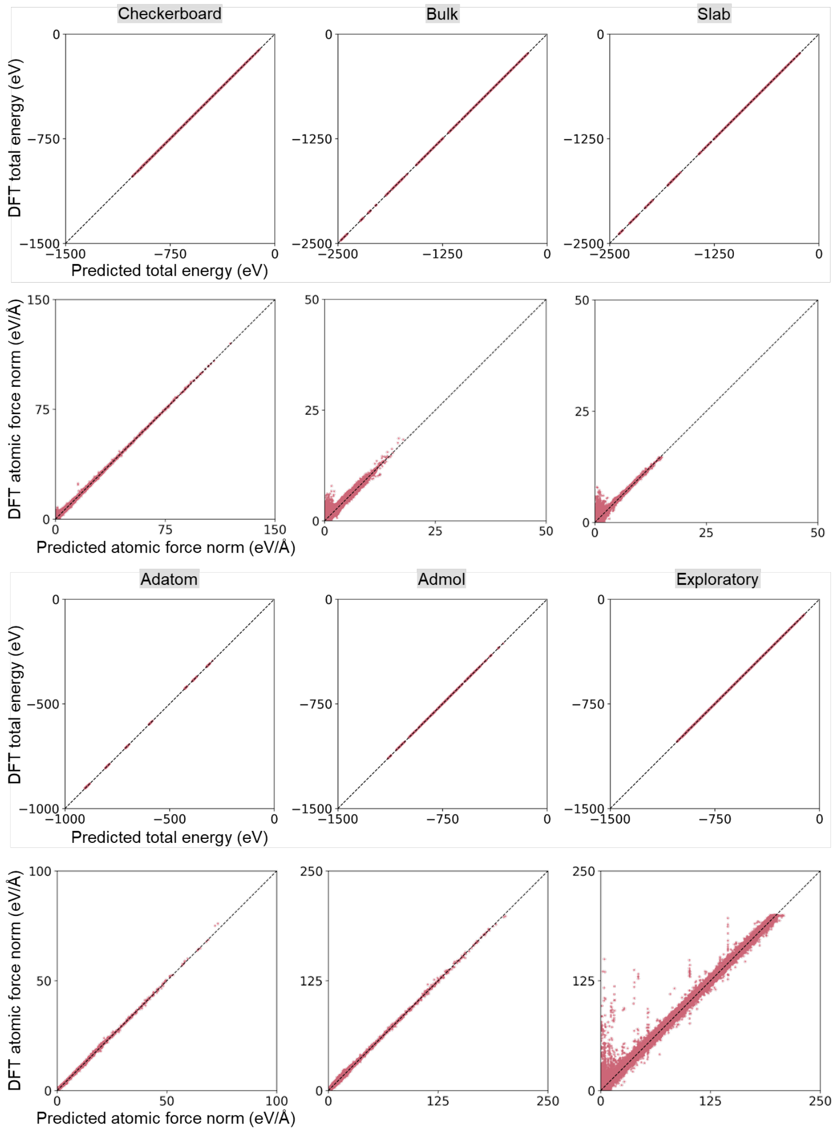
c

Validation sub-dataset of Titan25(G)	Training dataset	Per-atom energy error (meV/atom)		Atomic force error (meV/Å)	
		MAE	RMSE	MAE	RMSE
Checkerboard	Titan25(G)	3	4	88	144
	Titan25(G+I)	4	5	96	154
Bulk	Titan25(G)	2	2	67	107
	Titan25(G+I)	2	2	72	116
Slab	Titan25(G)	2	3	60	104
	Titan25(G+I)	3	3	65	111
Adatom	Titan25(G)	2	3	44	89
	Titan25(G+I)	3	4	46	92
Surface	Titan25(G)	3	5	142	347
	Titan25(G+I)	4	6	143	348
Exploratory	Titan25(G)	9	89	161	331
	Titan25(G+I)	10	89	160	328

Supplementary Fig. 4: **a**,**b**, Validation and training losses on Titan25(G) and Titan25(G+I), respectively. Learning rates are plotted together over training steps. **c**, Per-atom energy and atomic force errors of validation structures, separated by builder. The validation data of Titan25(G) were used in common for both trained models.



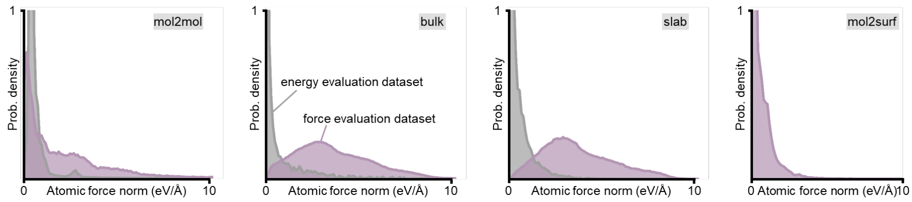
Supplementary Fig. 5: Parity plots of predictions of the MLIP, trained on Titan25(G), against DFT total energies (top) and atomic forces (bottom). They are grouped into six pairs according to their originating builders. The validation data of Titan25(G) were used for the plots.



Supplementary Fig. 5: (continued) Parity plots of predictions of the SNet-T25 model, trained on Titan25(G+I), against DFT total energies (top) and atomic forces (bottom).

9.5 GAIA-Bench

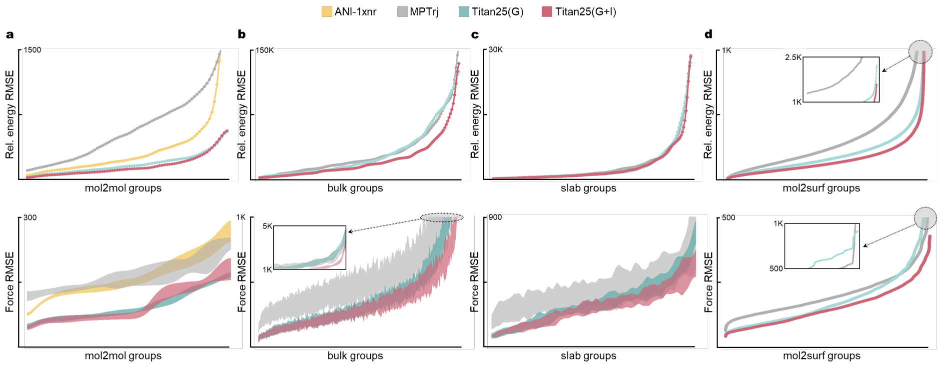
a



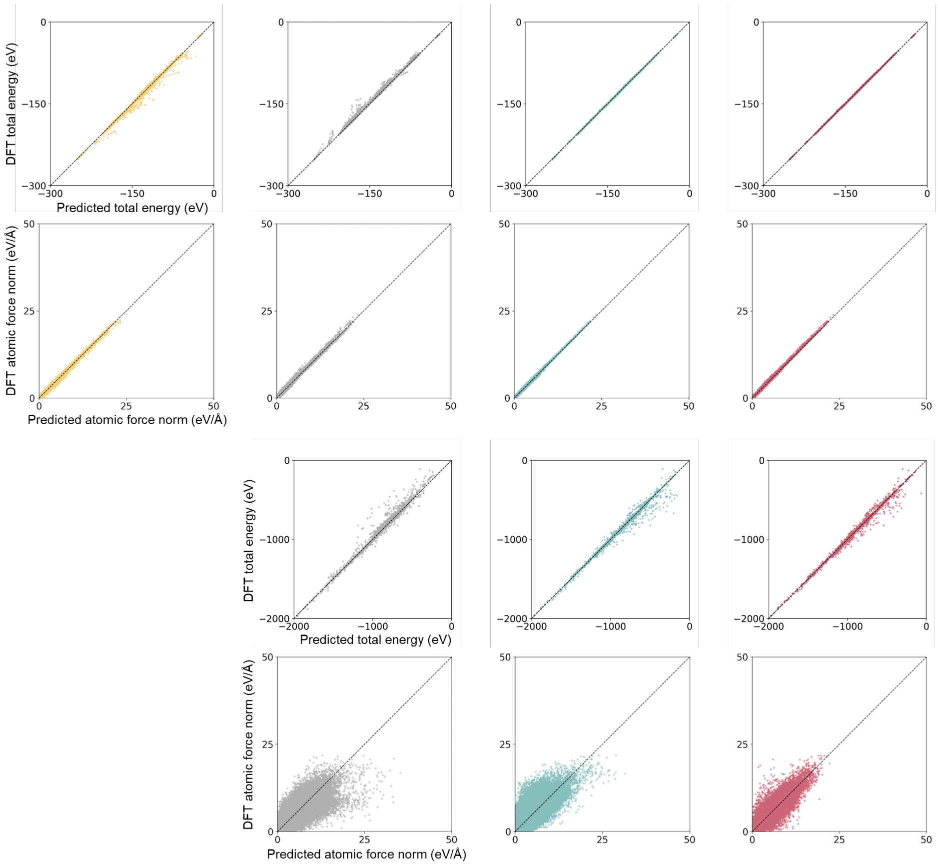
b

Benchmark task	Training dataset	Relative total energy error (meV)		Atomic force error (meV/Å or radian)			
		MAE	RMSE	MAE	RMSE	Magnitude error	Angle error
mol2mol	ANI-1xnr	115	560	105	169	117	0.25
	MPTrj	209	713	115	168	136	0.33
	Titan25(G)	82	211	65	**100**	**77**	0.18
	Titan25(G+I)	**61**	**195**	**64**	105	**77**	**0.17**
bulk	MPTrj	16.6K	43.1K	454	747	645	0.19
	Titan25(G)	19.6K	42.0K	362	746	460	0.18
	Titan25(G+I)	**14.4K**	**33.6K**	**284**	**511**	**363**	**0.14**
slab	MPTrj	2.0K	5.8K	297	437	418	0.13
	Titan25(G)	1.9K	5.8K	233	348	301	0.12
	Titan25(G+I)	**1.8K**	**5.4K**	**207**	**302**	**272**	**0.10**
mol2surf	MPTrj	239	503	131	236	176	0.74
	Titan25(G)	141	245	121	221	158	0.66
	Titan25(G+I)	**108**	**189**	**101**	**169**	**129**	**0.61**

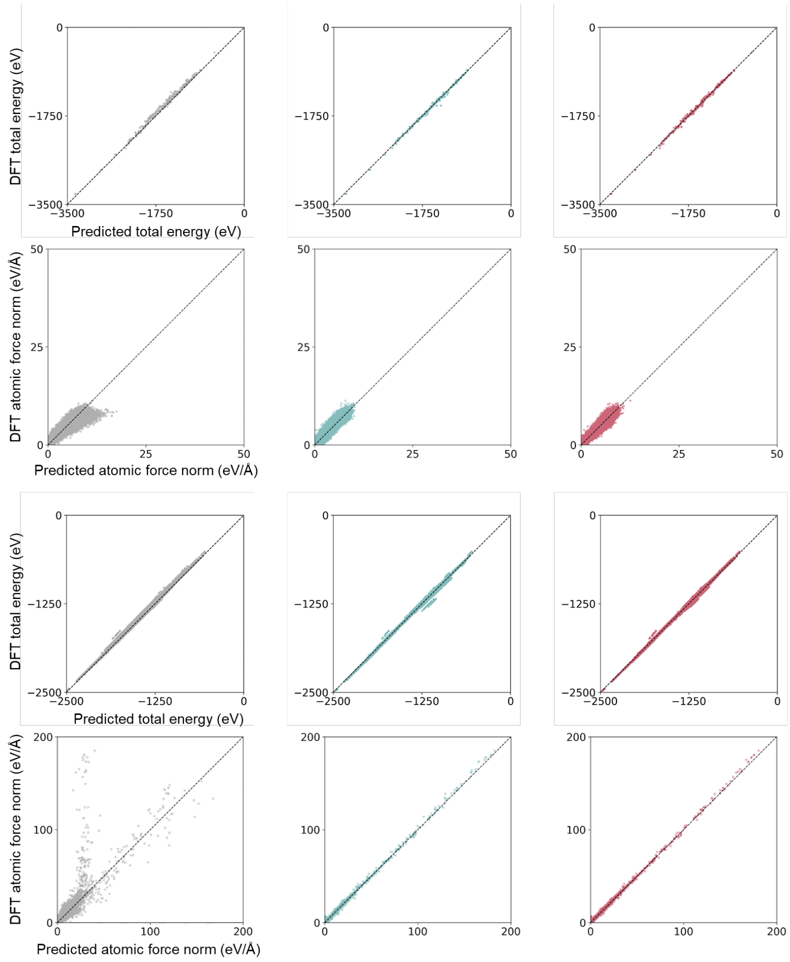
Supplementary Fig. 6: **a**, Probability density distributions of the GAIA-Bench benchmark tasks with respect to atomic force norm. Configurations sampled for force evaluations exhibit a greater weight at larger force norms compared to those sampled for energy evaluations, as designed. **b**, MAE and RMSE of relative energies and atomic forces on the four benchmark tasks. For forces, the averaged errors of the force magnitude and degree of angles (radian) are also reported. The bold characters indicate the lowest error on each benchmark task.



Supplementary Fig. 7: **a–d**, Error distributions of relative energies (top) and forces (bottom) for the four GAIA-Bench tasks, as in Fig. 3 but with RMSE.



Supplementary Fig. 8: Parity plots of GAIA-Bench comparing MLIP predictions with reference DFT total energies and atomic force norms. Each column includes the results of the MLIP trained on ANI-1xnr (yellow), MPTrj (gray), Titan25(G) (blue), and Titan25(G+I) (red), respectively. For ANI-1xnr, results are reported only for mol2mol, as its elemental coverage lacks the metallic species required for the other benchmark tasks. Rows correspond to the mol2mol and bulk benchmark results, each shown as a pair of plots for total energy (top) and atomic force norm (bottom).



Supplementary Fig. 8: (continued) Rows correspond to the slab and mol2surf benchmark results, each shown as a pair of plots for total energy (top) and atomic force norm (bottom).

Supplementary Table 2: MAE and RMSE of atomic forces, as in Supplementary Fig. 6b, but using the energy-evaluated configurations in this test. Boldface highlights the lowest error on each benchmark task.

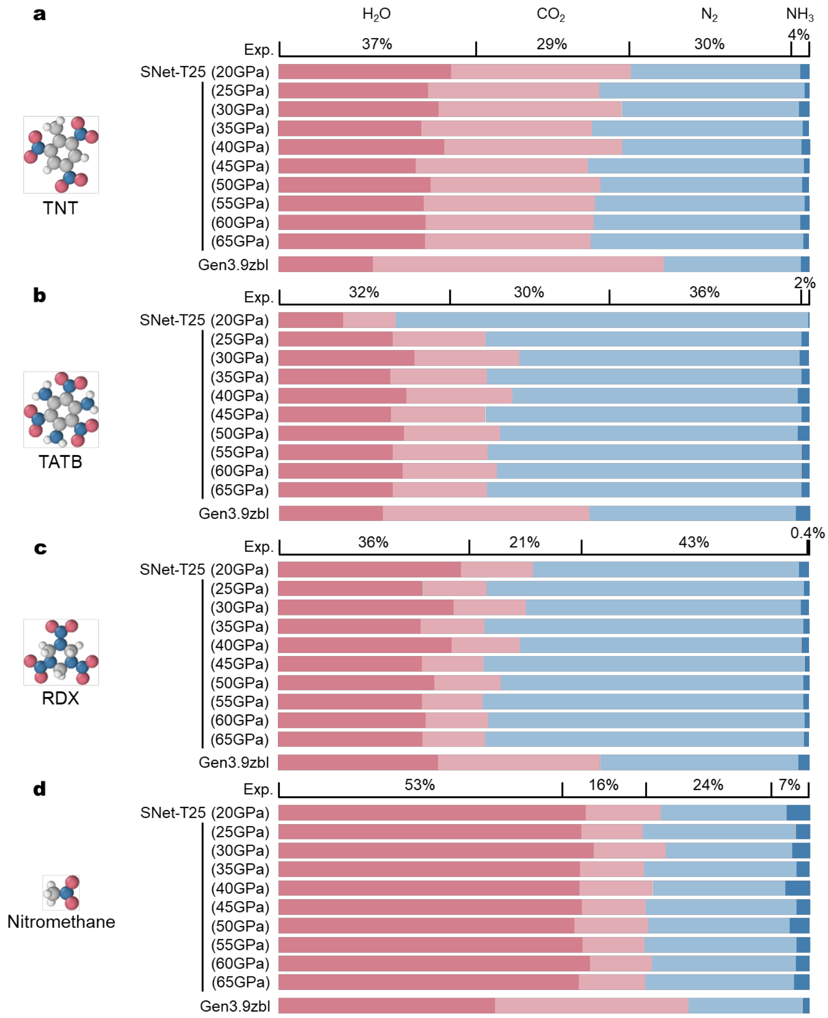
Benchmark task	Training dataset	Atomic force error (meV/Å or radian)			
		MAE	RMSE	Magnitude error	Angle error
mol2mol	ANI-1xnr	198	1.6K	274	0.36
	MPTrj	210	1.3K	299	0.47
	Titan25(G)	84	238	**98**	0.22
	Titan25(G+I)	**82**	**219**	99	**0.21**
bulk	MPTrj	109	410	196	**1.14**
	Titan25(G)	130	503	223	1.17
	Titan25(G+I)	**92**	**304**	**160**	**1.14**
slab	MPTrj	131	219	176	0.82
	Titan25(G)	145	263	189	0.80
	Titan25(G+I)	**119**	**195**	**152**	**0.76**
mol2surf	MPTrj	131	236	176	0.74
	Titan25(G)	121	221	158	0.66
	Titan25(G+I)	**101**	**169**	**129**	**0.61**

9.6 Test performance on public datasets

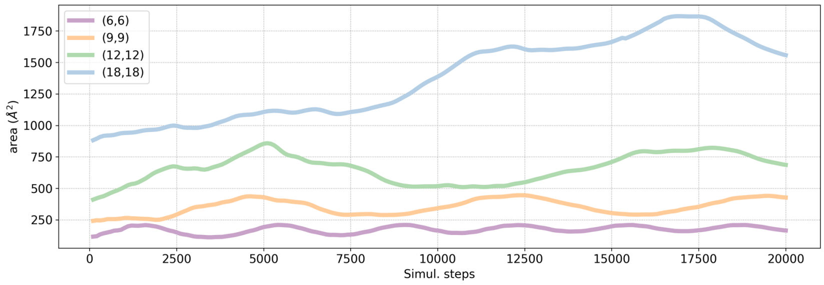
Supplementary Table 3: Energy and atomic force error comparison of four MLIP models on public test datasets: ANI-1x [40], GDB-13 [41], ANI-1xnr [19], and MPTrj [3]. All datasets were re-labeled with the PBE-D3 functional for consistency in the comparison. For the test datasets, ANI-1x and GDB-13 were evaluated by sampling 20000 data points randomly. In contrast, ANI-1xnr and MPTrj were partitioned into training (80%), validation (10%), and test (10%), with the model performance assessed on the held-out 10% validation sets to allow fair comparison with models trained on the corresponding training sets. MPTrj further restricted to configurations containing only elements present in Titan25. Entries corresponding to the lowest errors in each test dataset are highlighted in bold.

Test dataset	Training dataset	Relative total energy error (meV)		Atomic force error (meV/Å)	
		MAE	RMSE	MAE	RMSE
ANI-1x	ANI-1xnr	50	56	230	363
	MPTrj	35	44	290	363
	Titan25(G)	**27**	**31**	143	226
	Titan25(G+I)	**27**	32	**137**	**217**
GDB-13	ANI-1xnr	36	41	186	336
	MPTrj	**17**	24	190	319
	Titan25(G)	23	25	108	182
	Titan25(G+I)	20	**23**	**101**	**167**
ANI-1xnr	ANI-1xnr	4	6	139	251
	MPTrj	54	68	489	817
	Titan25(G)	13	15	243	373
	Titan25(G+I)	12	14	232	354
MPTrj	MPTrj	38	77	96	248
	Titan25(G)	107	174	132	272
	Titan25(G+I)	83	126	112	234

9.7 MLIP-MD evaluation



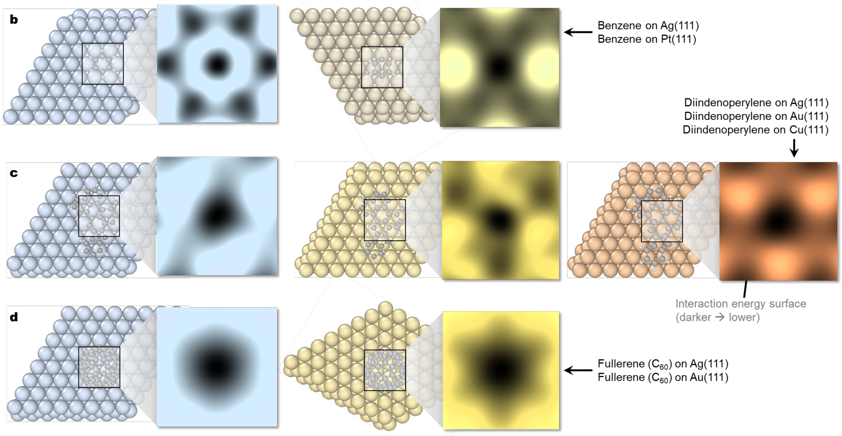
Supplementary Fig. 9: Normalized product ratio of the four energetic molecules simulated with SNet-T25 model. Gray, white, blue, and red spheres represent C, H, N, and O atoms, respectively. Results for Gen3.9zbl are taken from ref. [42].



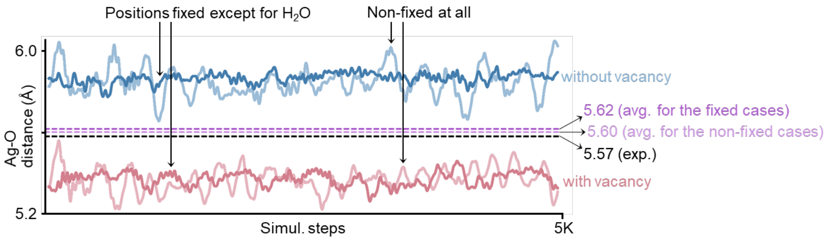
Supplementary Fig. 10: Projected area of CNTs with chirality indices (6,6), (9,9), (12,12), and (18,18) plotted against simulation steps. The fluctuations in cross-sectional area, observed in each curve, arise from structural transformations of the CNTs induced by thermal effects.

a

	Exp.	DFT (PBE+vdW^{surf})	SNet-T25	UMA-OMat	UMA-ODAC	UMA-OC20
Benzene on Ag(111)	3.04±0.02	3.00 (-0.04)	3.28 (+0.24)	3.18 (+0.14)	4.99 (+1.95)	5.08 (+2.04)
Benzene on Pt(111)	2.02±0.02	2.09 (+0.07)	2.09 (+0.07)	2.18 (+0.16)	2.27 (+0.25)	2.27 (+0.25)
Diindenoperylene on Ag(111)	3.01±0.04	2.94 (-0.07)	2.82 (-0.19)	2.95 (-0.06)	3.50 (+0.49)	3.89 (+0.88)
Diindenoperylene on Au(111)	3.17±0.03	3.22 (+0.05)	2.86 (-0.31)	2.96 (-0.21)	3.11 (-0.06)	4.17 (+1.00)
Diindenoperylene on Cu(111)	2.51±0.03	2.59 (+0.08)	2.64 (+0.13)	2.95 (+0.44)	2.61 (+0.10)	3.90 (+1.39)
Fullerene (C_{60}) on Ag(111)	2.01±0.10	1.99 (-0.02)	2.07 (+0.06)	2.07 (+0.06)	1.75 (-0.26)	3.95 (+1.94)
Fullerene (C_{60}) on Au(111)	1.80±0.10	1.81 (+0.01)	1.93 (+0.13)	1.64 (-0.16)	1.13 (-0.67)	3.61 (+1.81)



Supplementary Fig. 11: **a**, Vertical distances of π-conjugated molecules in their relaxed structures on various metallic surfaces, obtained with different MLIPs. The $\pm$ values indicate error bars for the experiments; the values in parentheses for the calculated results denote deviations from experiment. In the diindenoperylene rows, the trend of distance variations across the materials are described next to the values. SNet-T25 and UMA-OMat24 reproduce the experimental trend well. All quantities are reported in Å. **b**, Benzene on Ag(111) and Pt(111). **c**, Diindenoperylene on Ag(111), Au(111), and Cu(111). **d**, Fullerene on Ag(111) and Au(111). Note that there is a central vacancy on the metallic surface.



Supplementary Fig. 12: Evolution of the Ag–O distance during MD simulations of a fullerene on the Ag(111) surface. The red curves correspond to a fullerene encapsulating H_2O on Ag(111) with a central vacancy, whereas the blue curves represent the defect-free Ag(111) surface. Light-colored curves indicate unconstrained MD simulations; solid-colored curves denote simulations in which all atoms were fixed except for H_2O. The average Ag–O distances are shown as dashed lines in pale purple and purple, respectively.

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